

Next-Track Session Recommendation: Experiment Synthesis

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Abstract

This document is the living synthesis of the *next-track* sequence recommendation campaigns; the markdown reports in `experiments/` (filenames containing `nexttrack`) are the primary record. The task is next-item *retrieval*: given a listening *session* (an ordered sequence of tracks), rank the catalog and recover the next distinct track the user actually played, scored on a single leak-free chronological *next-distinct* split (7154 sessions / 19 402 items; 5723 train / 1431 test; cold-item rate 0.55). The headline metric is Recall@10, reported beside mean reciprocal rank (MRR) and the graded-relevance diagnostics artist@10 / genre@10 / artist_mrr. Twelve campaigns are consolidated. A session GRU alone only *ties* the first-order Markov co-listening bar (R@10 0.1069) on Recall@10 and loses on MRR, and neither loss-tuning nor any network topology (depth, width, residual, direction) beats it. The breakthrough is *rank fusion*: an R+M z-blend (GRU $\times$ Markov) beats the bar by $\sim +61\%$ Recall@10 (0.1726, three-seed confirmed), because the two legs rank *different* correct tracks; a learned XGB stacker underperforms the fixed z-blend and adding redundant neural legs does not help. The one genuinely additive new signal is *content*: an item-representation sweep found latent *dimension* (not the compressor) to be the lever, with a peak at PCA-192, and on that rich space a three-leg R+M+C z-blend (GRU $\times$ Markov $\times$ content-kNN) is the current best-on-record row at Recall@10 0.1859, MRR 0.1025. A subsequent literature-fit campaign then superseded that crown: a supervised *learned content projection* ($R'+M+C'$), which fits a linear metric map of the PCA-192 item space toward next-track adjacency by InfoNCE and retrieves the GRU and content legs in it, reaches Recall@10 0.2117, MRR 0.1220 (paired Δ over R+M+C +0.0259, CI above zero), is *seed-robust* across three seeds and *cold-reaching*, and hardens on a second disjoint 0.64-cold split (paired Δ +0.0426, CI above zero) — resolving the two thin caveats (determinism-only confirm, +0.0007 margin) that had bounded the R+M+C crown. Unlike every earlier ranking predictor it is registered, promoted and *servable* (`blend-gru-markov-content-proj`, best-models rank 1), closing the productization gap; an item-vector representation exploration and a continuous “sounds-alike” *music@10* metric corroborate the champion without dethroning it. A final measurement campaign adds a display-only *session-holisticness* metric suite (artist_adj@10 / mood_coh@10 / ild@10) that fingerprints the champion as the *most* artist-eager model measured (artist_adj@10 0.610) — with the eagerness sourced from its Markov bigram leg, not the RNN, and the GRU and LSTM statistically tied — and finds an eval-time MMR re-rank ($\lambda \approx 0.9$) a near-free holisticness lever while a training-time anti-eager regularizer is weak; nothing is promoted and the primary metric stays Recall@10. Two final architecture campaigns then close the remaining architecture axes, and both are nulls: a jointly-trained *dual-tower* GRU over paired causal views of the prefix fails to beat a parameter-matched single tower — the fourth independent learned-combiner null, and the one that answers the “train the fusion end-to-end” rebuttal — and a learned per-step *pre-encoder* in front of the recurrence is not merely useless but actively *harmful* in a single tower (-0.0140 Recall@10, CI below zero), a loss traced by a counterfactual interpretability read to half-wave rectification of a zero-centred latent deleting exactly its negative half (`frac_exact_zero` 0.5327 = `frac_negative_coords` 0.5327). A measured power analysis finally reframes all five architecture nulls: at $n = 1431$ the paired-bootstrap CI half-width is 0.0124 on Recall@10 while a whole architecture family spans only

0.028, so large harms were detected reliably while moderate wins were never detectable at all — and *holisticness@10* exists in three mutually non-comparable code generations, which is why this synthesis keeps its board denominated in Recall@10. A final measurement campaign then changes the *surface* rather than the model: the shipped product serves an *autoregressive walk*, while *holisticness@10* scores a one-shot top-10 whose *mood_coh* anchors to a *growing* prefix centroid, so journey drift was invisible by construction (and the composite is $5.5\times$ more sensitive to its relevance factor than to its eagerness factor). Four walk-level metrics calibrated against the user’s own listening (73 632 real consecutive transitions, median step cosine 0.261, 27.0% of them negative) show the shipped engine walking $2.5\times$ tighter than the listener it imitates (median step 0.661), and three deterministic retrieval-time controls — a seed anchor, a stride term, and a hard per-artist cap that *cannot be outscored* (two disjoint splits: $\Delta_{\text{artist_conc}} -0.527$ with $\Delta_{\text{music@10}} +0.026$, both CI clear of zero) — make a real-listening stride and a held vibe simultaneously reachable. On 40 identical held-out sessions under paired bootstrap the *dual-tower* family that three campaigns had refuted on recall is the winner on the stated objective (**latent/cummean**: stride error -0.2545 , genres $+1.50$, drift $+0.0351$, all CI clear of zero, vibe *held*), while pushing the single GRU or the champion to the same stride *costs* vibe — but **ground** straddles zero for every arm, so only journey *shape* is established, and the primary metric is unchanged.

1 Introduction

This document is the living synthesis of the next-track sequence-recommendation campaigns; the markdown reports in **experiments/** are the primary record. Each section transcribes (never re-computes) the quantitative results of a campaign, preserves its run and dataset identifiers verbatim for traceability, and adds the rationale and interpretation the terse reports omit. This instance was cloned from a sibling taste-fit / rotation lab and retargeted to a sequence ranking goal; the rotation campaigns are a separate line of work and are *not* covered here.

Task and split. The lab predicts, within a listening *session*, which track the user plays *next* — a next-item retrieval task. The model consumes the ordered session prefix and scores every catalog item; candidates are ranked and the held-out next *distinct* track is the target (a leave-last-out evaluation that excludes prefix items, matching the Markov parity convention). All work runs on one sequence artifact, **seq-20260713-014226** (7154 sessions / 19 402 items), split *chronologically by session start* into 5723 train / 1431 test at cut 2024-08-24. The defining property of this split is its **cold-item rate of 0.55**: more than half of test-session target tracks never appear in any training session. This is the leak-free forward reality, and it caps absolute recall by construction — every result is read against the incumbent baseline bar, never an absolute number.

Metrics. The headline metric is **Recall@10** (R@10, the fraction of test sessions whose held-out next track appears in the top ten retrieved), reported beside **mean reciprocal rank** (MRR, the average inverse rank of the true next track) and Recall@20. Since 2026-07-14 the evaluation also emits first-class *graded-relevance* metrics over the same prefix-excluded ranking (Section 7): **artist@k** (a top-*k* candidate shares the true next track’s artist), **genre@k** (likewise on primary genre), and **artist_mrr** (MRR of the first same-artist candidate). All confidence intervals are bootstrap 95% CIs over 2000 resamples of the 1431 test sessions, and the decisive statistic is the *paired* per-session Δ against a named reference. A board *win* requires beating the reference on *both* R@10 and MRR beyond non-overlapping marginal CIs (the strict bar), then a reproduction confirm; the paired- Δ CI-entirely-above-zero test is the more powerful alternative statistic (it crowned both champions) and can disagree with the strict bar. The graded metrics are the honest

scorecard: exact recall is modest by construction, but the champion lands the right *artist* in the top ten $\sim 32\%$ of the time and the right *genre* $\sim 40\%$ — far more useful than the $\sim 17\%$ exact number the cold-item rate imposes. Since 2026-07-20 the leaderboard also carries **music@10**, a continuous “sounds-alike” graded metric (Section 14): the mean best cosine, in a balanced intrinsic musical-distance space, between the true next track and the top-ten retrieved candidates — a smooth superset of Recall@10 that credits landing something that *sounds like* the truth on an exact miss. It is display-only; the primary metric stays Recall@10.

The standing champion. The current best-on-record *row* — and, since 2026-07-18, a *registered, promoted and servable* model — is a supervised *learned content projection*, $R' + M + C'$: a session GRU $\times$ a first-order Markov co-occurrence model $\times$ a content-kNN leg, where the GRU (R') and the content leg (C') retrieve in an InfoNCE metric map of the PCA-192 item space fit toward next-track adjacency, at R@10 0.2117, MRR 0.1220 (Sections 11–12). It supersedes the three-leg $R + M + C$ z-blend on the same PCA-192 space (R@10 0.1859; Section 10), which had itself dethroned the two-leg $R + M$ z-blend on the 64-dim song-autoencoder space (AE-64, R@10 0.1726; Section 6). The projection is *seed-robust* and *two-split hardened*, and it is deployable via `POST /api/models/blend-gru-markov-content-proj/predict` — closing the productization gap the earlier off-server fusion drivers left open. The prior champions remain useful references ($R + M$ is still the leanest two-leg blend), but they are no longer the deployable anchor.

2 Background: the sequence task, the baselines, and the metrics

This section gives the minimum theory for a reader who knows machine learning but not this repo. It is keyed to the project’s living facts file and grows only when a new predictor family or metric enters the reports.

The models and baselines in play.

Session GRU / LSTM (seq-nexttrack). A recurrent neural network that reads the ordered session prefix one item at a time (each item represented by its item-latent vector) and emits a predicted next-item latent; candidates are ranked by cosine similarity to that prediction. The GRU (gated recurrent unit) and LSTM (long short-term memory) are the two recurrent cells tried. The recurrence captures *session context* — the trajectory of the listening session — rather than a single previous item.

Training loss: InfoNCE versus cosine. The sequence model can be trained to regress the next-item latent by *cosine* loss (pull the prediction toward the true next latent) or by **InfoNCE**, a contrastive objective that pulls the prediction toward the true next item and pushes it away from a batch of negatives. InfoNCE is the single most important lever in the whole arc: it is $\approx 2.3\times$ cosine on Recall@10, because ranking retrieval rewards a representation that *separates* the true next item from distractors, which a contrastive loss optimizes directly and a pointwise regression does not.

First-order Markov co-occurrence (the bar, B2). A train-only transition model: a `track_bigram` counting “what track follows what” with artist/genre back-off for unseen pairs (the transition term the sibling app already ships). It directly *memorizes* adjacency, which is exactly what exact next-track ranking rewards, and it is the reference bar every arm is measured against. A second-order Markov (trigram) back-off was also tried

and is essentially neutral — sessions are too short/sparse for a trigram to fire, so it falls through to the bigram.

item2vec. A skip-gram (SGNS) co-listening embedding trained on the *train* sessions only (a leak-safe analogue of word2vec, where “words” are tracks and “sentences” are sessions); candidates are ranked by cosine to the mean-pooled prefix. On 5723 train sessions it is too sparse to localize and is the weakest leg on record; the full-history co-listening vectors are deliberately *not* used because they leak across the split.

content-kNN. A training-free retrieval leg: cosine of each candidate’s item latent to the prefix latents (max-aggregated). It is weak standalone on exact recall but captures *vibe* similarity the counts miss (its genre@10 already edges the Markov bar), and it is the one genuinely additive third leg for the fusion blend.

Rank-fusion z-blend versus a learned stacker. Two ways to combine legs. The **z-blend** normalizes each leg’s candidate scores (z-score) and takes a weighted sum ($\alpha \cdot z(\text{leg}) + \dots$) — a fixed, parameter-light fuser. The **XGB stacker** is a gradient-boosted meta-learner over per-candidate features (leg scores + popularity + recency + artist/genre-match), trained pointwise with sampled negatives. Across two campaigns the fixed z-blend *beats* the learned stacker decisively, because the stacker’s pointwise binary training does not optimize the ranking objective as well as directly fusing normalized scores.

Learned content projection (supervised metric map). A linear map W of the item-latent space fit by **InfoNCE** over the train consecutive pairs ($u \rightarrow v$) (self-loops dropped, Markov convention), so cosine in W -space reflects “likely to follow.” It is leak-free (train pairs + intrinsic content latents only, never the split). Both the content-kNN leg (C') and the GRU (R') then retrieve in the projected space; Markov is unchanged. It is the one *supervised* alternative to the unsupervised, variance-ordered PCA geometry, and — because content is the only channel with cold reach — it lifts *both* legs that share the space, making it the current champion (Section 11).

Dual-tower recurrent fusion (seq-dualgru). Two *independent* recurrent towers (separate weights) read the prefix in parallel over configurable strictly-causal *views* of it — **latent** (the raw item vector), **delta** (the step-to-step difference) or **cummean** (the running session centroid) — and their concatenated per-step hidden states are read out by a fusion head (**fusion_layers** = 0 for a pure linear read, = 1 for a 256-wide MLP). It reuses the **seq-nexttrack** objective, batching, early stopping and evaluation verbatim, so it isolates architecture as the only variable. The motivating idea is that two towers should help only if they see genuinely different information; Section 16 tests that and refutes it.

Per-step pre-encoder. A nonlinear re-embedding placed *in front of* the recurrence — [Linear, ReLU, Dropout] applied at each timestep, exposed as **pre_hidden/pre_layers**. It is an unusually sharp test object: a GRU’s own input transform is *already* linear, so a linear pre-encoder at width $\geq$ the input dimension is exactly as expressive as no pre-encoder at all, and the *rectifier* is the only change — which is what makes Section 17’s null mechanistically attributable rather than merely negative.

Item-latent spaces: song-AE versus PCA, and the wrong-collection trap. The sequence model predicts *into*, and retrieves *over*, a per-item latent space, so the choice of that space is a first-class axis. Three families appear:

AE-64 (`spotify_tracks_song_ae`). A 64-dim nonlinear *song-autoencoder* latent, the original champion item space. An autoencoder compresses each item by reconstructing it through a bottleneck; its equal-weight-per-block reconstruction loss spreads capacity evenly across content, acoustic and catalog blocks.

PCA- k (`spotify_tracks_song_pca $k$`). A *linear* principal components compression at dim $k \in \{32, 64, 128, 192, 256\}$, which orders dimensions by explained variance and decorrelates them. As Sections 8–9 show, this variance-ordered, decorrelated geometry is *easier* for the GRU to regress into than the AE’s entangled equal-weight latents, so PCA beats the AE at every $\text{dim} \geq 64$.

The wrong-collection trap. Both of the above are fit *only* from the leak-safe `spotify_tracks_content` source (intrinsic content/acoustic/catalog fields; no behavioral or target field). The default `spotify_tracks` 200-dim collection is a *behavioral* co-listening embedding that soft-leaks replay behavior and must **never** be used as an item space.

3 Next-track leaderboard

Table 1 is the standing best-on-record board. It is on the single leak-free next-distinct split and sorted by Recall@10. Since 2026-07-18 the sequence family carries a ranking serving contract, so the champion learned-projection blend is a *promoted, servable* model (`blend-gru-markov-content-proj`, best-models rank 1), not merely a board row (Section 12); the earlier blends below it, and the leg standalones, remain off-server references. The item spaces are all fit from the leak-safe content source (Section 2). Since 2026-07-22 the evaluation also carries a *display-only session-holisticness* suite (`artist_adj@10` / `mood_coh@10` / `ild@10`; Section 15); it does not reorder this board (the primary metric stays Recall@10), but it fingerprints the champion as the most artist-eager row. Since 2026-07-25 the board also carries three *refuted* architecture datapoints — the widest single-GRU capacity control, the best dual-tower arm and the best single-tower pre-encoder arm (Sections 16–17). They are listed because they *bound* their axes, not because they are candidates: none beats the single-GRU baseline by the paired statistic and none was registered. On the server-effective primary metric, and on why this board stays denominated in Recall@10 across three holisticness generations, see Section 18.2.

4 The sequence sweep: a GRU-infonce ties the Markov bar

Rationale. The instance’s opening campaign asked the founding question: does a session-context sequence model — a GRU/LSTM over the ordered prefix, predicting the next track’s 64-dim song-AE latent and retrieving by cosine — beat the first-order Markov transition model the sibling app already ships, *forward*, on the leak-free session split? Four arms crossed the recurrent cell (GRU, LSTM), capacity (`hidden` 128, 256) and, decisively, the training loss (cosine versus InfoNCE), at 60 epochs.

Results. No arm beat the bar. The best arm (M3, GRU `hidden128` InfoNCE) reached a statistical *tie* with the Markov bar on R@10 (0.0894, paired- Δ CI includes zero) but *lost* on MRR (0.0388 versus 0.0694) and Recall@20. The dominant lever was the loss objective: InfoNCE was $\approx 2.3\times$ cosine on R@10 (0.0894 versus 0.0391), while capacity (`h256`) and the LSTM cell barely moved the cosine arms (all ~ 0.038 – 0.041). The popularity and recency floors sat near zero (R@10 0.0007).

Table 1: Next-track best-on-record board, on the leak-free next-distinct split (1431 test sessions; bootstrap 95% CI over 2000 resamples). R@10, MRR, artist@10, genre@10, music@10 higher-better; the first-order Markov model is the incumbent reference bar. music@10 values are the offline backfill (Section 14); rows without a mapped run are left blank. Champion row and prior champions bolded; best cell per metric bolded. The three refuted architecture rows added 2026-07-25/26 are shown to *bound* their axes and are excluded from the best-cell bolding; they carry no marginal CI, only a paired Δ against the single-GRU baseline (Tables 15 and 17).

Model	R@10 [95% CI]	MRR	art@10	gen@10	mus@10	Note
<i>R'+M+C'</i> , PCA-192 (champion, servable)	0.2117 [0.191, 0.233]	0.1220	0.333	0.401	0.454	new champion
R+M+C, PCA-192 (prior fusion champ.)	0.1859 [0.166, 0.207]	0.1025	0.331	0.394	0.449	content additive
R+M+C, PCA-128	0.1824 [0.162, 0.203]	0.0993	0.332	0.396	—	content additive
R+M , AE-64 (prior champion)	0.1726 [0.154, 0.192]	0.0963	0.322	0.397	0.441	prod. anchor; 3-seed
R+M+C, AE-64 (graded-preferred)	0.1740 [0.154, 0.194]	0.1014	0.330	0.400	—	content only tied
R+M, PCA-192	0.1712 [0.151, 0.192]	0.0952	0.329	0.398	0.443	$\approx$ prior champion
R+M, PCA-128	0.1670 [0.148, 0.187]	0.0920	0.329	0.398	0.444	GRU gain not carried
R+A+M, AE-64	0.1684	0.0943	—	—	—	ANN leg redundant
LSTM single, PCA-128	0.1272 [0.110, 0.145]	0.0495	0.335	0.458	0.451	best single; $\ll$ blend
GRU single h454, PCA-192	0.1251	0.0539	0.338	0.454	0.457	width closed
GRU single, PCA-192	0.1230 [0.106, 0.140]	0.0534	0.338	0.450	0.456	dim point-optimum
dual-tower GRU f0, PCA-192	0.1188	0.0563	0.337	0.456	0.455	dual tower: refuted
GRU single, PCA-128 (base)	0.1146 [0.099, 0.132]	0.0489	0.333	0.452	0.457	confirmed base space
GRU single, PCA-256	0.1167 [0.101, 0.134]	0.0502	0.338	0.458	0.456	past the dim peak
pre-MLP $\rightarrow$ GRU, PCA-192	0.1090	0.0446	0.324	0.469	0.450	pre-encoder harmful
B2 first-order Markov (bar)	0.1069 [0.092, 0.124]	0.0694	0.298	0.389	0.407	app incumbent
GRU single, AE-64 (control)	0.0957 [0.081, 0.111]	0.0411	0.294	0.450	0.432	bake-off control
ANN pooled-MLP, PCA-128	0.0901 [0.076, 0.105]	0.0394	0.299	0.435	0.426	loses to the GRU
XGB stacker (M+G / M+G+A)	0.080–0.087	0.042–0.049	0.31	0.38	0.41	combiner underperform
content-kNN (song-AE)	0.0636 [0.050, 0.077]	0.0321	0.272	0.405	—	the additive C leg
item2vec (train-only SGNS)	0.0517	0.0214	0.166	0.312	—	drags the blend
B0 popularity / B1 recency	0.0007	—	—	—	0.185	floors

Interpretation. A latent-plus-single-cosine-hop is a harder route to the *exact* next track than a co-listening bigram: the bigram directly memorizes “what follows what,” while the GRU must land near the right point in a 64-d content space and then retrieve, losing on precise ranking (MRR). The finding is consistent with a taste-drift reality — 0.55 cold items, identity/adjacency-driven behavior — and it set the agenda: InfoNCE is the base loss, and the natural next arm is a *blend* of the GRU’s candidate scores with the Markov bigram. *Source: 2026-07-13-nexttrack-sequence-sweep.md.*

5 Topology scan: depth, width and direction are a spent lever

Rationale. With the loss fixed at InfoNCE, this scan isolated network *topology*: does depth (stacked layers), width (hidden 256), an inter-layer residual stack or bidirectionality lift the sequence model over the Markov bar? Because a bidirectional RNN under per-step teacher forcing would leak future items, the bidirectional arm used a leak-free last-item objective (encode the full prefix, predict only the held-out next item).

Results. Refuted. No arm beat the bar on both metrics. The best arm (width h256, R@10 0.0957) is a *tie* on R@10 and loses on MRR (0.0411 versus 0.0694). Depth is flat 1 $\rightarrow$ 2 layers (0.0894 $\rightarrow$ 0.0922, in-band) and *hurts* at 3 layers (0.0825, out-of-band below the bar); the residual stack over depth-2/h256 added nothing. The bidirectional arm collapsed (0.0482), but this is an

objective artifact, not a directionality verdict: the leak-free last-item objective supplies ~ 1 training target per session versus teacher forcing’s ~ 13 next-step pairs, so it trained on $\sim 13\times$ less signal.

Interpretation. Architecture is not the lever for next-track on this corpus. The transferable latent-retrieval signal caps at $R@10 \sim 0.09\text{--}0.10$ across depth, width, residual and direction, below the co-listening bigram — consistent with the whole arc that the loss objective moved the needle and capacity/topology do not. The one remaining unexplored lever was the candidate framing / fusion, which the next campaign pulled decisively. *Source: 2026-07-13-nexttrack-topology-scan.md.*

6 The GRU $\times$ Markov fusion champion, and why more ensembling fails

Rationale. The sweep and the topology scan agreed that the best single GRU only *ties* the Markov bar on $R@10$ while losing on MRR. This campaign asked whether *combining* models beats the bar, and how far ensembling goes. Three legs were defined, each emitting full-vocab candidate scores: **M** = the train-only Markov bigram (the bar), **R** = the GRU-inforce-h256 recurrence, and **A** = an ANN bag-of-prefix MLP (a feed-forward twin of R). They were combined by a weighted z-blend and, as a learned alternative, an XGB stacker.

Results. The **R+M z-blend** ($\alpha=0.5$) *beats the bar decisively*: $R@10$ 0.1726 [0.154, 0.192], MRR 0.0963, a paired Δ over Markov of +0.0657 [+0.049, +0.083] (CI entirely above zero) — $\sim +61\%$ $R@10$, +39% MRR — on a robust plateau across $\alpha \in [0.25, 0.75]$ and reproduced across three seeds (0.1705 / 0.1726 / 0.1726). It was registered as the **seq-blend** predictor and the **blend-gru-markov** definition, the first next-track champion. Further ensembling then *failed* to improve on it (Table 2): R+A+M (0.1684) is slightly worse, R+A ($\approx$ GRU alone, 0.0894) is much worse, and both XGB stacker variants (M+R 0.1020; M+R+A 0.0887) barely clear Markov alone.

Table 2: The fusion campaign: the R+M z-blend beats the Markov bar decisively, and no richer ensemble improves on it. $R@10$ and MRR higher-better; champion bolded.

Model	$R@10$ [95% CI]	MRR	Note
R+M z-blend $\alpha=0.5$ (champion)	0.1726 [0.154, 0.192]	0.0963	Δ vs Markov +0.066 [+0.05, +0.08]
R+A+M z-blend	0.1684	0.0943	ANN leg redundant with GRU
B2 Markov bar (M)	0.1069 [0.092, 0.123]	0.0694	the reference
XGB stacker M+R	0.1020	0.0561	learned combiner $\ll$ z-blend
GRU alone (R)	0.0957	0.0411	ties the bar on $R@10$ only
R+A z-blend	0.0894	0.0377	$\approx$ GRU alone
XGB stacker M+R+A	0.0887	0.0572	barely clears Markov

Interpretation. The win is about signal *diversity*, not model count. The GRU (latent similarity) and Markov (co-occurrence counts) rank *different* correct next-tracks, so fusing them roughly doubles the hit rate; piling on more of the *same* kind of signal — another neural latent predictor (A), redundant with R — adds nothing. The learned XGB stacker *underperforms* the simple z-blend, counter to the usual “stacking beats blending” intuition: its pointwise binary-with-negatives training does not optimize the ranking objective as well as directly fusing normalized scores. The remaining headroom is bounded by the 0.55 cold-item rate and requires *genuinely new signal*, not more model fusion — the hypothesis the next campaign tests. *Source: 2026-07-14-nexttrack-blend-and-ensemble.md.*

7 Graded-relevance metrics and new base learners: content is the only additive signal

Rationale. Two threads in one campaign. (A) Make the evaluation credit landing the right *band/vibe* even when the exact track is wrong — the observation that “the first choice is usually not the exact track, but the artist is usually right” — by promoting *artist@k*, *genre@k* and *artist_mrr* to first-class metrics wired end-to-end (`seq_common.eval_from_scores` → Rust Metrics / MetricsSpec → `domain.toml` → UI). (B) Try genuinely new leak-free base learners — **C** = content-kNN, **M2** = a second-order Markov trigram with back-off, and **I** = a train-only item2vec SGNS — scored on both the exact and the new graded metrics.

Results. The exact-metric path was byte-identical (purely additive keys); the champion reproduced R@10 0.1726 / MRR 0.0963 exactly. No new base learner beat the bar standalone (M2 0.1076 ≈ M 0.1069 — the trigram falls through to the bigram; item2vec 0.0517 is the weakest leg). Blended into the champion, **content is the only additive signal: R+M+C** (adding the content leg) is the top blend on MRR (0.0963→0.1014), artist@10 (0.3215→0.3298) and artist_mrr, and ties the champion on exact R@10 (paired Δ +0.0014, CI straddles zero). Item2vec *drags* the blend (R+M+I 0.1349); the trigram is neutral.

Interpretation. Content-kNN ranks *different* near-correct tracks (latent vibe similarity) than the GRU (session recurrence) and Markov (co-occurrence), exactly the diversity the fusion campaign predicted new signal would need. On the AE-64 space the content leg only *tied* the champion on exact recall while lifting the graded metrics, so it was carried forward as the graded-preferred third leg, not a new exact champion. That “tie” becomes a *win* once the item space is richer (Section 10): on PCA-192 the same R+M+C construction is the strongest on record. Co-listening item2vec and the trigram Markov added nothing on this data. *Source: 2026-07-14-nexttrack-graded-relevance-and-new-legs.md.*

8 The item-representation compression bake-off

Rationale. The champion’s item space — a 64-dim AE latent — was one arbitrary early choice, and the sequence GRU predicts *into* and retrieves *over* whatever geometry that space imposes. This campaign asks which *compression* is the best space for the GRU: linear PCA versus the nonlinear autoencoder, each at latent dim 32 / 64 / 128. Six alternative spaces were baked from the *same* leak-safe 384-dim content source (`spotify_tracks_content`), each fit unsupervised over all 23 529 items using only intrinsic content/acoustic/catalog fields, then scored under a *frozen* GRU readout (`gru-infonce-h256`), so the item latent space is the only axis that moves. The design bet was that the AE’s equal-weight-per-block reconstruction loss would *preserve* the small acoustic/catalog blocks that PCA’s text-dominated global variance would starve; the null was that dim, not the compressor, sets the ceiling.

Results. The parity gate passed: the AE-64 control reproduced its on-record R@10 0.0957. All six spaces trained cleanly (Table 3; Figure 1). **PCA-128** is the single-GRU winner at R@10 0.1146, a paired Δ of +0.0189 [+0.0042, +0.0335] over the AE-64 control — a CI *entirely above zero*, the only arm that clears the control significantly — and it also tops MRR, artist@10 and artist_mrr. PCA-64 is the runner-up (0.1041, tie). The two 32-dim spaces sit far below the 64-dim control (both CIs below zero). The training-free content-kNN probe is flat at ~0.06 across all six spaces and does *not* track the GRU ranking.

The single-GRU win does **not** survive into the production blend: re-fusing the winning PCA-128 GRU leg with the Markov leg scores R@10 0.1670, *below* the AE-64 champion blend (0.1726)

Table 3: The item-representation compression bake-off: six item spaces (linear PCA vs the nonlinear AE $\times$ latent dim 32 / 64 / 128), each scored under the *same* frozen **gru-infonce-h256** readout (1431 test sessions; paired- Δ vs the in-batch AE-64 control). R@10, MRR, artist@10, genre@10 higher-better. Winner (PCA-128) bolded; best cell per metric bolded.

Space	Run id	R@10 [95% CI]	Δ vs AE-64	MRR	art@10	gen@10
PCA-128 (winner)	run-2026071 5-042100-a1 d9d	0.1146 [0.099 , 0.132]	+0.019 [+0.004, +0.034]	0.0489	0.3326	0.4521
PCA-64 (runner-up)	run-2026071 5-042100-88 51d	0.1041 [0.088, 0.121]	+0.008 (tie)	0.0416	0.3117	0.4598
AE-64 (control)	run-2026071 5-042059-dd 261	0.0957 [0.081, 0.111]	— (control)	0.0411	0.2942	0.4500
AE-128	run-2026071 5-042059-e9 6ae	0.0929 [0.078, 0.108]	−0.003 (tie)	0.0398	0.3026	0.4640
AE-32	run-2026071 5-042059-d3 f9e	0.0545 [0.043, 0.066]	−0.041	0.0274	0.2376	0.3955
PCA-32	run-2026071 5-042059-b1 242	0.0412 [0.031, 0.052]	−0.055	0.0224	0.2187	0.3816

with overlapping CIs. Nothing was promoted; PCA-128 was carried forward only as the sanctioned Phase-2 base space.

Interpretation. Latent dimensionality is the dominant lever and the compressor is secondary. The design’s acoustic-preservation hypothesis is *refuted*: at generous dim the linear PCA beats the nonlinear AE, and although AE-32 preserves the numeric/acoustic blocks near-perfectly it ranks near the bottom. What helps is a *variance-ordered, decorrelated* target geometry that the GRU finds easier to regress into than the AE’s entangled equal-weight latents. The AE’s edge at dim 32 is a purely low-capacity phenomenon that reverses once dim is generous. The flat content-kNN probe that fails to track the GRU ranking is a flagged divergence: the win comes from the *learnability* of PCA’s target geometry, not a demonstrably richer space. And the load-bearing caution is the blend result: a clearly stronger GRU leg bought *nothing* at the blend, because the Markov leg is representation-independent and the extra tracks the better GRU recovers overlap what Markov already supplies. *Source: 2026-07-15-nexttrack-compression-representation-bakeoff.md.*

9 The dimension ceiling: PCA peaks at 192

Rationale. With dimension established as the lever and PCA still climbing at 128, this leg extended the PCA sweep upward (PCA-192, PCA-256) to locate the real optimum before Phase 2 committed a base space, and added AE-192 to test whether $\text{PCA} \geq \text{AE}$ holds at higher dimension. The protocol was identical (same frozen **gru-infonce-h256** readout, same parity gate); the reference is PCA-128 (run-20260715-042100-a1d9d).

Results. The PCA R@10-versus-dimension curve **peaks at ~ 192** (R@10 0.1230) and *turns over* by 256 (0.1167); it was not still climbing. The full trend is $32 \rightarrow 0.041$, $64 \rightarrow 0.104$, $128 \rightarrow 0.115$, **$192 \rightarrow 0.123$** , $256 \rightarrow 0.117$ (Figure 2). PCA-192 is the point-optimum, but its paired Δ versus PCA-128 is +0.0084 [−0.0007, +0.0189] — it misses the “CI entirely above zero” bar by 0.0007,

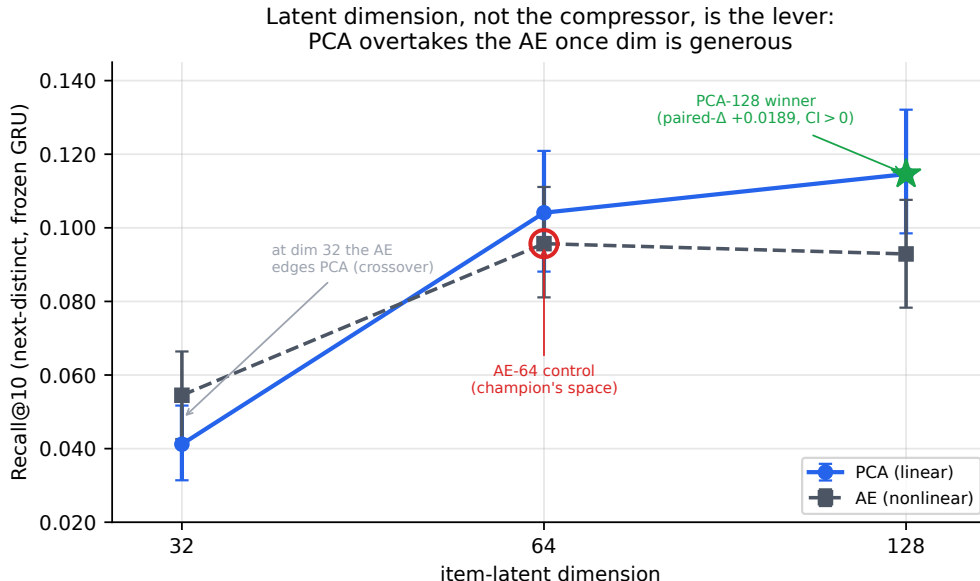


Figure 1: Latent dimension, not the compressor, is the lever. Both series climb steeply from dim 32 to 64; PCA (blue) then keeps climbing to 128 and *overtakes* the AE, while the AE (grey) plateaus — its 128-dim point actually dips below its own 64-dim control. At dim 32 the AE edges PCA, a low-capacity crossover that reverses once dim is generous. The AE-64 control (red ring) is the champion’s item space; the PCA-128 winner (green star) is the only arm whose paired- Δ CI clears zero. Error bars are bootstrap 95% CIs; the neighbouring CIs overlap, so the win rests on the *paired* test.

a whisker — so **PCA-128 stands as the Phase-2 base**, with PCA-192 flagged as the confirm-and-promote candidate. $\text{PCA} \geq \text{AE}$ holds strongly and the gap is widest at 192 (PCA-192 0.123 versus AE-192 0.098); the AE plateaus flat at $\sim 0.09\text{--}0.10$ from dim 64 upward.

Interpretation. Extra variance past ~ 192 (PCA-256 captures cumulative explained-variance ratio 0.997) is low-signal / noise dimensions the GRU cannot exploit and that slightly dilute the target geometry. The AE’s flat plateau reconfirms the refuted acoustic-preservation hypothesis at higher dimension: its equal-weight reconstruction objective keeps spending capacity on the tiny numeric/acoustic blocks (reconstruction $R^2 \approx 0.999$ at 192) rather than the high-variance text/catalog structure the GRU needs. As before, the training-free content-kNN probe stays flat and does not reproduce the GRU’s peak, so the dimension optimum is about target *learnability*, not intrinsic retrieval richness. *Source: 2026-07-15b-nexttrack-compression-dim-extension.md.*

10 The model-family sweep and the new champion, R+M+C on PCA-192

Rationale. With the compression axis settled (PCA-128 base, PCA-192 candidate), this campaign swept the model *family* on the chosen spaces: does a different recurrent cell (LSTM), a non-recurrent pooled MLP (ANN), or a *learned* XGB stacker beat the fixed GRU $\times$ Markov z-blend — and does adding the content-kNN third leg (R+M+C) on the *rich* PCA space finally beat the champion? A frozen held config ran across all arms; only the model family and the representation varied, and the server was serialized (`--max-runs 1`) to avoid a thread-oversubscription throughput

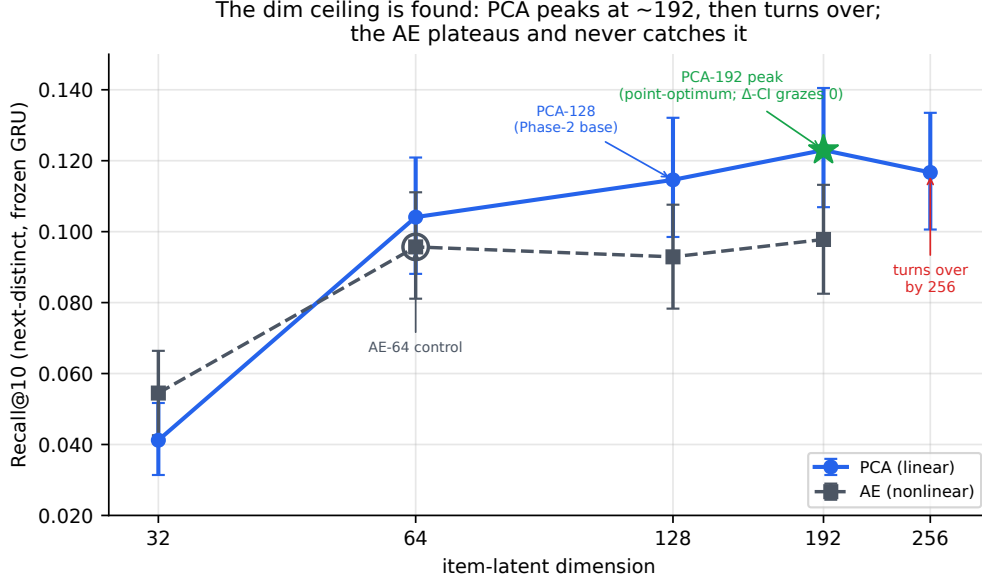


Figure 2: The dimension ceiling is found. Extending the compression sweep of Figure 1 to PCA-192/256 and AE-192, the PCA curve (blue) peaks at ~ 192 ($R@10$ 0.123) and turns over by 256, while the AE (grey) plateaus flat at ~ 0.09 – 0.10 and never catches it — the $PCA \geq AE$ gap is widest at 192. PCA-192 is the point-optimum (green star) but its paired Δ over PCA-128 grazes zero, so PCA-128 stays the Phase-2 single-model base. Error bars are bootstrap 95% CIs over the 1431 test sessions.

trap.

Results. A **new best-on-record row**: the **R+M+C z-blend on PCA-192** ($GRU \times$ train-only Markov $\times$ content-kNN, equal thirds) reaches $R@10$ 0.1859 [0.166, 0.207], MRR 0.1025, a paired Δ over the prior champion of +0.0133 [+0.0007, +0.0259] — entirely above zero, with no MRR regression (Table 4; Figure 3). The content leg is genuinely additive on the rich space, beating same-space R+M by +0.0154 (PCA-128) / +0.0147 (PCA-192), both CIs above zero — *stronger* than the AE-64 R+M+C that only tied (Section 7), because the richer, higher-EVR space makes the content leg pay off. The learned **XGB stacker badly underperforms** the fixed z-blend (0.080–0.087, losing even to its own single GRU base leg); the **LSTM is the best single model** (PCA-128 0.1272, paired Δ over the GRU +0.0126, CI above zero) but far below the blends; and the Phase-2 single base **stays PCA-128**.

The crown, and its two honest caveats. The board-event candidate was confirmed by reproducing the GRU-192 leg five times (the original plus four reruns) and recomputing R+M+C each time; the paired Δ over the prior champion was +0.0133 [+0.0007, +0.0259] *every* rerun, with the MRR lead intact, so R+M+C on PCA-192 was recorded as the new champion row. Two caveats travel with the crown and must not be omitted. First, **the reproduction spread was exactly zero** because `seq-nexttrack` training is deterministic on this CPU — all five GRU-192 checkpoints were byte-identical — so the confirm verified *determinism, not seed-robustness*; the only uncertainty quantified is the single-split bootstrap. Second, **the paired Δ lower bound is +0.0007**: the crown clears zero by a whisker on a single split. The crown is real by the pre-agreed criterion, but the margin over the prior champion is thin, and hardening it needs a second leak-free test split (or seed-varied legs once the blend is generalized to three legs).

Interpretation. The compression-plus-model-family thread is otherwise closed: PCA-128 is the

Table 4: Phase-2 model-family sweep on the rich PCA spaces, ranked by R@10 (higher-better). The fixed z-blend is the combiner of record; the learned XGB stacker underperforms it badly. Champion row and prior champion bolded; best cell per metric bolded. Paired Δ is per-session against the named reference.

Model	Run id / source	R@10 [95% CI]	MRR	paired Δ
R+M+C, PCA-192 (champ. row)	analyze_rmc (A4)	0.1859 [0.166 , 0.207]	0.1025	vs champ +0.0133 [+0.001, +0.026]
R+M+C, PCA-128	analyze_rmc (A2)	0.1824 [0.162, 0.203]	0.0993	vs champ +0.0098 (tie)
R+M, AE-64 (prior champion)	run-20260714- 170928-645ae	0.1726 [0.154, 0.192]	0.0963	— (reference)
R+M, PCA-192	run-20260715- 142440-96f53	0.1712 [0.151, 0.192]	0.0952	vs champ -0.0014 (tie)
R+M, PCA-128	run-20260715- 044920-36dce	0.1670 [0.148, 0.187]	0.0920	vs champ -0.0056 (tie)
LSTM single, PCA-128	run-20260715- 142440-546d0	0.1272 [0.110, 0.145]	0.0495	vs GRU-128 +0.0126 (> 0)
GRU single, PCA-192	run-20260715- 142440-89d6a	0.1230 [0.106, 0.140]	0.0534	vs GRU-128 +0.0084 (tie)
ANN pooled-MLP, PCA-128	run-20260715- 142440-7e528	0.0901 [0.076, 0.105]	0.0394	vs GRU-128 -0.0245
XGB stacker M+G, PCA-128	run-20260715- 142440-371b1	0.0867 [0.073, 0.101]	0.0493	vs GRU-128 -0.0280
XGB stacker M+G+A, PCA-192	run-20260715- 142440-f111d	0.0853 [0.071, 0.099]	0.0490	vs GRU-192 -0.0377

single-model base, the LSTM is the best single model, the learned stacker is dead as a combiner, and the fixed z-blend is the combiner of record. The load-bearing result is that *content is the one leg that adds genuinely new ranking signal, and it pays off more on the higher-EVR PCA space* — the exception to the standing rule that single-model / representation gains do not carry to the blend (the R+M pair on the rich PCA spaces only ties the AE-64 champion). The productization gap remains: the champion is train-only and off-server, so the prior champion R+M on AE-64 stays the deployable anchor until the three-leg blend is registered with a `predict` subcommand. *Source: 2026-07-15c-nexttrack-phase2-model-sweep.md.*

11 The learned content projection: a new champion from the literature-fit campaign

Rationale. The R+M+C champion (Section 10) carried two thin caveats — a determinism-only reproduction confirm and a paired- Δ lower bound of only +0.0007 — and its headroom was capped by the 0.55 cold-item rate, which only *new* signal can attack. Guided by a survey of the established session-recommendation literature (the Jannach–Ludewig session-kNN line, sequential rules / factorized Markov, GRU4Rec training tricks, rank fusion and learning-to-rank, content / cold-start-aware session rec), this campaign pre-registered three tiers of classic methods and asked which actually improve the fit to *this* problem — a single-user, session-based, next-distinct *retrieval* task with 55% cold targets. All arms were measured off-server on the PCA-192 champion artifact `seq-20260715-131139`, reusing the sanctioned driver that reproduces the champion (R+M+C 0.1859) exactly; the decision statistic is the paired per-session Δ against that champion.

Results — the one win. A supervised **learned content projection** is a new best-on-record

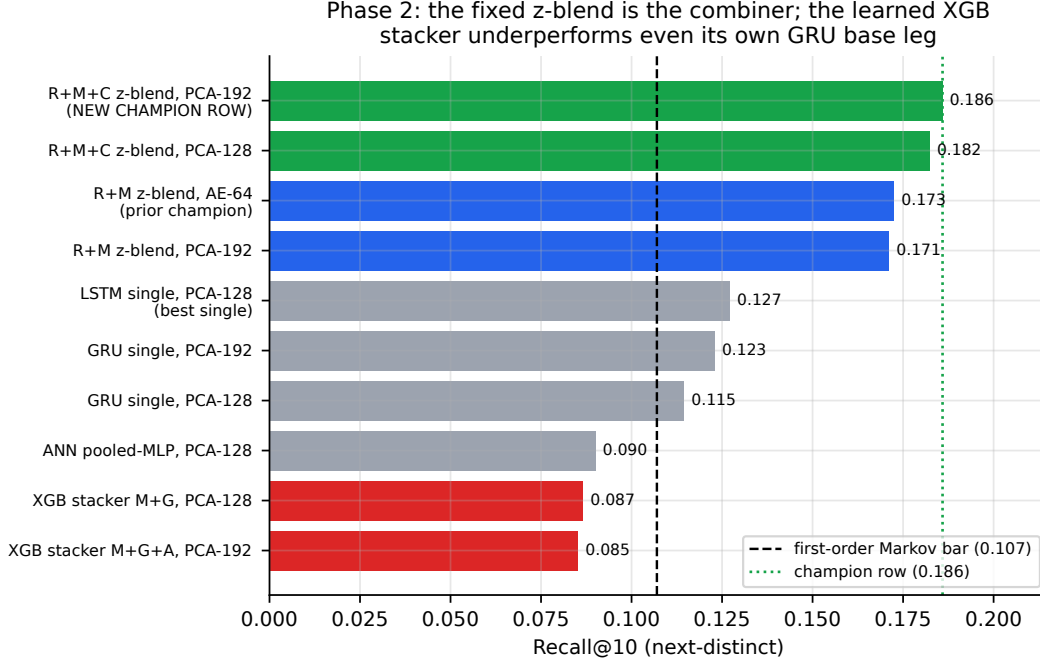


Figure 3: The Phase-2 model-family sweep. The fixed z-blend is the combiner of record: R+M+C on PCA-192 (green) is the new champion row, the content leg is genuinely additive on the rich space, and the learned XGB stacker (red) badly underperforms — it loses even to its own single GRU base leg. LSTM edges GRU as the best single model (grey) but sits far below the blends. The first-order Markov bar (0.107) is the incumbent reference; the champion row is at 0.186.

row. Fitting a linear metric map W of the PCA-192 content space toward next-track adjacency (InfoNCE over train consecutive pairs, self-loops dropped) and then retrieving *both* the content leg (C') and the GRU (R') in that projected space gives $R'+M+C'z\text{-blend } R@10$ 0.2117 [0.191, 0.233], **MRR** 0.1220 — a paired Δ over the champion of +0.0259 [+0.012, +0.041], entirely above zero (Table 5). Unlike the R+M+C crown it is **seed-robust** (three seeds; paired Δ +0.026 / +0.025 / +0.029, every CI > 0 — a genuine seed-varied confirm, since the projection has random init and shuffled batches) and **cold-reaching**: the projected content leg C' scores cold targets 0.152 $\approx$ warm 0.151, versus Markov’s warm 0.226 / cold 0.010, so it reaches never-seen items as well as seen ones — genuine content generalization, the one channel that reaches the 55% cold majority. The symmetric single map beats the directional two-tower, and rank-192 = rank-64 (the gain is the supervision, not the capacity). As a side effect the projected single GRU R' (~ 0.176) nearly matches the *old* champion blend and crushes the prior best single (LSTM 0.127).

The lanes that closed. Every other literature-fit direction failed to beat the fixed z-blend, exactly as the survey predicted for a single-user, cold-heavy corpus (Table 6). Principled Markov smoothing (absolute-discounting + a Kneser–Ney continuation floor) is a real but tiny standalone estimator gain (0.1069 $\rightarrow$ 0.1104) that washes out in the blend (-0.0021 , CI straddles 0 — the content leg already covers the marginal cold mass it recovers). A ranking-aware combiner has no honest headroom: even the *oracle* weights, fit directly on the test labels, reach only 0.1901 (+0.0042, straddles 0), and RRF rank fusion *hurts* recall (-0.029 , CI<0, discarding the score magnitude the z-blend keeps). A gated V-SKNN session-kNN is warm-only against 55% cold targets and *drags* the blend (-0.038 , CI<0) — the single-user setting strips the cross-user col-

Table 5: Tier A1 of the literature-fit campaign: the learned content projection on PCA-192, versus the R+M+C champion (1431 test sessions; bootstrap 95% CI; paired Δ per session vs the champion). Winner ($R'+M+C'$) bolded; best cell per metric bolded.

Model	R@10 [95% CI]	MRR	art@10	gen@10	paired Δ vs champion
R+M+C champion (thirds)	0.1859 [0.166, 0.207]	0.1025	0.331	0.394	— (reference)
$R'+M+C'$ (both projected)	0.2117 [0.191, 0.233]	0.1220	0.333	0.401	+0.0259 [+0.012, +0.041] > 0
R+M+C+C' (4-leg)	0.2061 [0.185, 0.227]	0.1148	0.345	0.410	+0.0203 [+0.013, +0.029] > 0
R+M+C' (swap C $\rightarrow$ C')	0.1971 [0.177, 0.218]	0.1107	0.334	0.400	+0.0112 [−0.000, +0.023] cand.
C' proj-content (sym, r192)	0.1516	0.0708	0.305	0.401	vs unsup. C 0.0685: +0.083
C' dir. two-tower (r192)	0.1349	0.0552	0.291	0.393	symmetric wins

laborative neighborhood that drives its multi-user benchmark wins. And GRU4Rec importance negatives (a larger support-weighted negative bank) give a within-noise standalone tick (+0.005) that also washes out (−0.0014, tie). **The only additive gain comes from new signal, not smarter fusion.**

Table 6: The literature-fit lanes that closed: every borrowed method fails to beat the fixed z-blend champion (1431 test sessions; blend paired Δ per session vs the champion).

Lane	Standalone R@10	Blend Δ vs champion	Verdict
Markov KN-smoothing (Tier A2)	0.1069 $\rightarrow$ 0.1104	−0.0021 (straddles 0)	null in blend
Ranking combiner, <i>oracle</i> (B1)	0.1901 (blend)	+0.0042 (straddles 0)	no honest headroom
RRF rank fusion (B1)	0.1572 (blend)	−0.029 (CI<0)	hurts recall
V-SKNN session-kNN (B2)	0.0671	−0.038 (CI<0)	drags (warm-only)
GRU importance negatives (C)	0.1230 $\rightarrow$ 0.1279	−0.0014 (tie)	washes out

Interpretation. The supervised content projection is the first genuinely-additive new signal since the content leg itself, and it works for the reason the survey flagged: content is the only channel with cold reach, and replacing the unsupervised, variance-ordered PCA geometry with a target-fit one lifts *both* legs that share it. The flashy borrowed methods — session-kNN, learned combiners, GRU-side loss tricks — are dead ends here because they either re-shuffle the same three legs (no headroom over the z-blend, even for the oracle) or import a collaborative assumption the single-user corpus violates. The one movable secondary axis is graded relevance: the oracle / RRF reweights and the projection nudge artist@10 / genre@10, a lever the exact-recall axis does not share. *Source: 2026-07-18-nexttrack-literature-fit-campaign.md.*

12 Registration, the objective / rank / τ scan, and second-split hardening

Rationale. The projection win landed as an off-server driver — the same productization gap the R+M+C crown had. This follow-through implemented the projection as a first-class capability of the seq-blend predictor (hyperparameters `projection`, `projection_rank`, `projection_objective`, `projection_tau`, ...: it fits the metric map from train consecutive pairs, saves it as `projection.npz` in the run dir, and bakes it into the promoted model for serving), then (1) reproduced the win on-server and registered + promoted it, and (2) ran a pre-approved objective / rank / τ ablation scan to confirm the default is the robust config to promote.

Registration. The verify run (`run-20260718-191539-1b89c-seq-blend`, definition default = In-

foNCE / full-rank / τ 0.07) reproduced the off-server driver to four decimals — R@10 0.21174, MRR 0.12199 — and was promoted via the API as `blend-gru-markov-content-proj`, the projection matrix baked into the model directory for serving. GET `/api/best-models` confirms it at **rank 1** (0.21174), above the prior servable champion `blend-gru-markov-content` (rank 2, 0.18588). With the sequence family’s ranking serving contract this is a genuinely deployable model, servable via POST `/api/models/blend-gru-markov-content-proj/predict`.

The scan. The decision rule — re-promote only a config beating the default on *both* R@10 and MRR by more than the $\sim\pm 0.015$ noise band — confirms the default (Table 7). **InfoNCE** $\gg$ **BPR** is the sharpest axis: both BPR cells (0.1579, 0.1523) fall ~ 0.055 below the InfoNCE default and land *below even the old R+M+C champion* (0.1859), so the pairwise BPR map produces a materially worse geometry for both shared-latent legs. **Full-rank** $\geq$ **rank-64** (-0.006 , a tie — reconfirming “the gain is the supervision, not the capacity”). And τ is **mildly sensitive and monotone up** over the tested range (0.05 $\rightarrow$ 0.1929, 0.07 $\rightarrow$ 0.2117, 0.10 $\rightarrow$ 0.2229) — but the nominal τ -0.10 top cell beats the default by only $+0.0112$ on recall, *inside* the noise band, so it is a tie and the default **HELD**. Unlike the reweighting axis of Section 11, the graded metrics here *track* recall (best and worst together), because the scan moves the *signal quality* rather than re-fusing fixed legs.

Table 7: The on-server objective / rank / τ ablation scan (6 cells, sorted by recall@10). All overlay the promoted definition; hit@10 = recall@10 by construction. Promoted champion (default, held) bolded; best cell per metric bolded (all held by the sub-noise τ -0.10 cell).

Config	recall@10	MRR	art@10	gen@10	run id
S2 infonce / full / τ 0.10	0.2229	0.1273	0.3424	0.4032	run-20260718-193111-a781c
infonce / full / τ 0.07 (promoted, held)	0.2117	0.1220	0.3326	0.4011	run-20260718-191539-1b89c
S3 infonce / rank 64 / τ 0.07	0.2055	0.1169	0.3326	0.3990	run-20260718-193650-66c96
S1 infonce / full / τ 0.05	0.1929	0.1111	0.3354	0.3997	run-20260718-192443-57432
S4 bpr / full / τ 0.07	0.1579	0.0939	0.3187	0.3913	run-20260718-194228-4cc06
S5 bpr / rank 64 / τ 0.07	0.1523	0.0909	0.3194	0.3906	run-20260718-194832-d070f

12.1 Second-split hardening

Rationale. The projection crown was measured on a single leak-free split. This validation-only pass re-ran it — plus the two reference blends — on a **second, disjoint, leak-free split** to settle whether the win is split-specific. Nothing is promoted or re-promoted. The split `seq-20260718-211238` is an *earlier-holdout* 60–80% chronological band on the same PCA-192 space (train 4,292 [earliest 60%] / test 1,431 [60–80% band]); the item space is byte-identical to the champion split, and the test window is **disjoint in both index and time** from the champion split’s 80–100% test. Its **cold-item rate** is 0.64 (versus 0.55), a stronger, independent cold-reaching stress test — so absolute R@10 lands *below* the first-split 0.212 by construction, and the verdict is the paired per-session Δ on *this* split, not absolute reproduction.

Verdict — crown hardened. The decision rule (paired Δ vs R+M+C with 95% CI entirely > 0 and MRR not regressing) is met: paired Δ $+0.0426$ [$+0.0280, +0.0573$], MRR $+0.0234$ (no

Table 8: Second-split hardening on the disjoint 0.64-cold earlier-holdout split `seq-20260718-211238` (1431 test sessions; validation-only, nothing promoted). Point metrics (top) and the paired per-session Δ (bottom, 2000-resample bootstrap, seed 1337). Projection champion bolded.

Config	recall@10	MRR	art@10	gen@10	run id
$R'+M+C'$ projection champion	0.1705	0.1050	0.2991	0.3417	run-20260718-212639-50e1d
R+M+C crown (primary reference)	0.1279	0.0816	0.2956	0.3410	run-20260718-213156-cf1dd
R+M 2-leg (context reference)	0.1174	0.0791	0.2844	0.3368	run-20260718-214114-93bad
<i>Paired Δ recall@10 (projection – reference):</i>					
PROJ vs R+M+C (primary)	+0.0426	[+0.0280, +0.0573] — entirely > 0; MRR +0.0234 (no regression)			
PROJ vs R+M (context)	+0.0531	[+0.0391, +0.0678] — entirely > 0			

regression). The projection’s advantage is not split-specific — in fact the margin is *wider* here (+0.0426) than on the first split (+0.0259), consistent with the cold-reaching mechanism (Figure 4). Combined with the prior three-seed robustness and the cold-reach decomposition, the crown is now hardened on two disjoint leak-free splits — resolving both caveats that travelled with the R+M+C crown. *Source: 2026-07-18b-nexttrack-projection-registration-and-scan.md.*

13 Item-vector representation: is the geometry the bottleneck?

Rationale. The compression axis (Sections 8–9) found latent *dimension* to dominate but never asked whether the *composition* of the 192-d item vector — the equal-weighted 4-block PCA (text / numeric / acoustic / categorical) — is itself a handicap. An explained-variance-ratio (EVR) preflight said it might be: the raw input variance budget is 44% acoustic + 43% numeric versus only 2.5% text (text is L2-normalized, so its per-dimension scale is tiny), so full-matrix PCA-192 spends **87% of its variance budget** reconstructing the numeric + acoustic blocks to $R^2 = 1.000$ while the signal-carrying text block reaches only $R^2 = 0.717$; per-column standardize (whitening) flips the captured budget to 73% text / 21% categorical (text R^2 0.905) (Figure 5). Eight alternative item-latent spaces (whitening $\times$ acoustic-drop, text-only, text+categorical, and the musical-distance metric geometry) were built parity-gated to the champion split (vocab / split byte-identical; only `item_latents.f32` differs) and each evaluated under *two* readouts: **(A)** the champion projection `blend-gru-markov-content-proj` (the InfoNCE map re-fits on that space — the space’s ceiling under the winning recipe) and **(C)** the frozen `gru-infonce-h256` single (raw representation quality). Validation-only; nothing promoted.

Verdict — exact recall. **Readout A: no variant beats the PCA-192 champion with paired Δ CI entirely > 0.** The two nominally-positive spaces (V3 std-noaco +0.0049, V5 balanced-pca192 +0.0042) straddle zero; two spaces are significantly *worse* (V4 textcat −0.0168, dropping numeric+acoustic costs the projection real signal; V6 sonic-64 −0.0182, the expected 64-d handicap). **Readout C: six of eight variants beat the frozen GRU with paired Δ CI entirely > 0** (+0.025 to +0.036; best V5 balanced +0.0356, V3 std-noaco +0.0342). So the raw full-matrix PCA-192 geometry *is* a handicap — suppressing the loud acoustic / numeric directions lifts the raw GRU by ~ 20 –29% relative — **but the gain is fully absorbed by the projection:** whitening lifts readout C, not readout A, because the learned InfoNCE map already re-weights away those directions from data. Whitening and the learned projection are the *same fix*, and the champion has already pulled it (Figure 6).

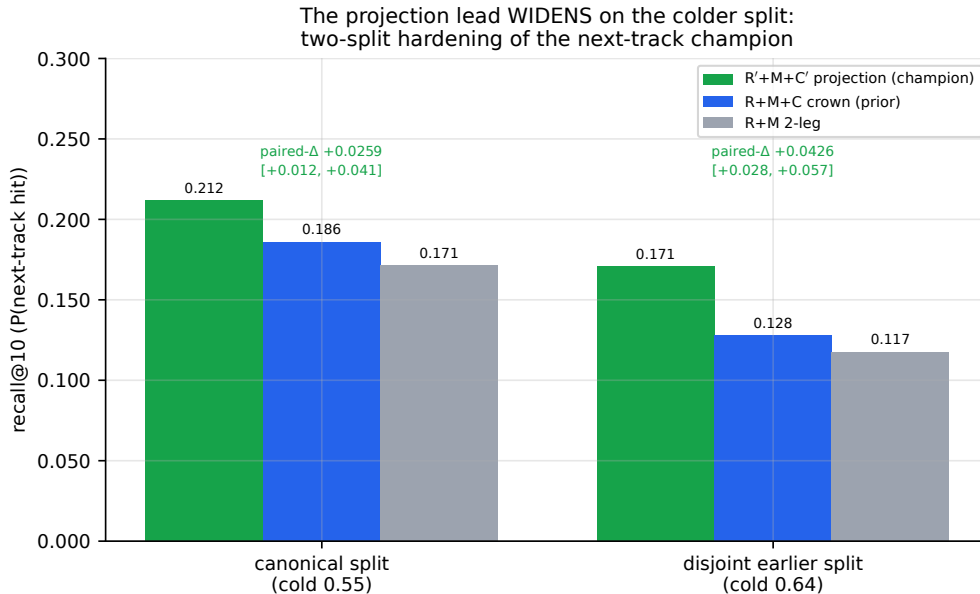


Figure 4: The projection lead *widens* on the colder split. On the canonical 0.55-cold split the projection champion beats the R+M+C crown by paired Δ +0.0259; on a disjoint, colder 0.64-cold earlier-holdout split the margin *grows* to +0.0426 (both CIs entirely > 0). Absolute R@10 is lower on the colder split by construction (more targets are cold); the verdict is the paired Δ , which is unambiguous on both. The mechanism: on a colder split the Markov / content-kNN legs of the crown reach fewer targets, and the projection’s cold-reaching C' leg carries proportionally more of the win.

Graded and music@10. The representation levers move the *graded* axis more cleanly than exact recall: whitening lifts the frozen GRU’s genre@10 to ~ 0.47 (V7 std-txtcatup 0.4717, V3 std-noaco 0.4710 vs baseline 0.4500), and under the projection a text-only space (V10) posts the readout-A maxima on artist@10 (0.3466) and genre@10 (0.4144) while trading -0.0119 exact recall. The continuous **music@10** metric (Section 14) *corroborates* the exact-recall verdict rather than overturning it: the projection champion leads on it (0.4538) over the R+M+C crown (0.4491), and under the champion projection it is flat across the eight representations (~ 0.447 – 0.460). Re-basing the geometry moves band / vibe matching, but exact next-track adjacency at the champion ceiling is immovable — which is precisely the axis the supervised projection owns.

Interpretation. The equal-weighted PCA-192 item vector is a genuine handicap for a *raw* model, and the learned projection is exactly the right response to it — this campaign independently validates *why* the projection is the champion (it is the representation fix, learned) and confirms there is no free lunch left in re-basing the input geometry. The one caution: dropping numeric+acoustic *entirely* (V4 textcat) hurts the champion even though it helps the raw GRU, so the numeric block carries something the projection uses — keep numeric, whiten / drop only acoustic. The most promising open direction is composing the two fixes: fitting the projection *on* a whitened / std-noaco / metric space to see whether it reaches further or is redundant. *Source: 2026-07-18c-nexttrack-representation-exploration.md.*

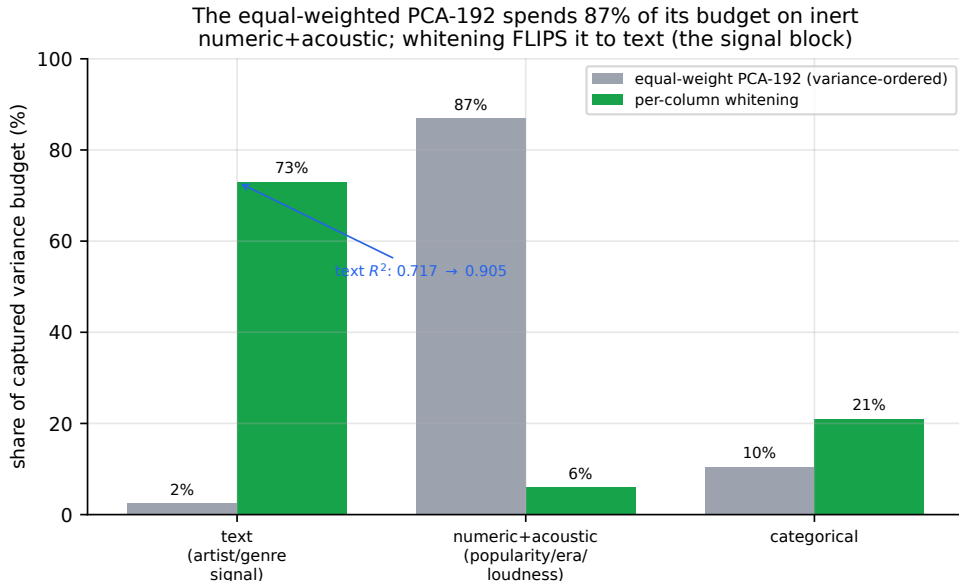


Figure 5: Why the item vector is a handicap for a raw model — and why the projection is the fix. The equal-weighted PCA-192 orders dimensions by raw variance, so 87% of its captured budget reconstructs the behaviorally-inert numeric + acoustic blocks (popularity / era / loudness) while the signal-carrying text block (2.5% of raw input variance, L2-normalized) is under-preserved (R^2 0.717). Per-column whitening flips the captured budget to 73% text / 21% categorical (text R^2 0.905). The learned InfoNCE projection learns this same reweighting from data.

14 The music@10 metric: the recall champion is not the musically-closest

Rationale. Exact Recall@10 is unforgiving on a 55%-cold single-user corpus, and the artist@10 / genre@10 diagnostics are coarse category hits. This campaign introduces a *continuous* “sounds-alike” graded metric, **music@10**, and back-fills it across every next-track run. It is defined as the mean over test cases of the best (max) cosine, in a *balanced* intrinsic musical-distance space (spotify_tracks_content_metric: a block-weighted, partial-whitened fusion of the whitened 64-d song-AE latent, the 384-d name/tag text embedding, five metadata numerics, and a per-genre AE prototype), between the true next track and the top-ten retrieved candidates. A top-ten that contains the exact truth scores 1.0, so music@10 is a *smooth superset* of Recall@10 — it credits landing something that *sounds like* the truth even on an exact / artist / genre miss. It is display-only; the primary metric stays Recall@10.

Results. music@10 was recomputed offline and non-destructively across all 53 next-track runs (from each run’s stored predictions.json and its dataset items.json, on the shared 1431-case split). It **correlates** with Recall@10 ($r = 0.609$) — a better ranker also lands closer misses — but it is far from redundant, and it crowns a *different* winner (Table 10; Figure 7). The Recall@10 champion is a **seq-blend** run (0.223), but the *musically-closest* predictions come from **seq-nexttrack** — which regresses the next track’s content latent directly and retrieves by cosine — at best music@10 0.482 and the highest family mean (0.448). When a next-track model misses the exact track, it still tends to surface something that *sounds* right; the **seq-popularity** floor is correctly worst (music@10 0.185), confirming the metric is not trivially saturated.

Table 9: Item-vector representation exploration: eight alternative 192-d item spaces vs the PCA-192 champion, under readout A (champion projection) and readout C (frozen GRU). Paired Δ on exact hit@10 (2000-resample bootstrap, rng 1337) vs the same-readout baseline; a lift needs the CI entirely > 0 . Baseline rows and the best readout-C lift (V5) bolded.

Variant (space)	dataset id	Readout A (projection)		Readout C (frozen GRU)	
		R@10	Δ hit@10 [CI]	R@10	Δ hit@10 [CI]
full — PCA-192 (baseline)	seq-20260715-131139	0.2117	(baseline)	0.1230	(baseline)
V1 noaco	seq-20260718-222742	0.2089	-0.0028 [-0.013, +0.008]	0.1481	+0.0252 [+0.009, +0.041]
V2 std (whiten all)	seq-20260718-222754	0.2082	-0.0035 [-0.015, +0.008]	0.1530	+0.0300 [+0.011, +0.049]
V3 std-noaco	seq-20260718-222807	0.2166	+0.0049 [-0.008, +0.017]	0.1572	+0.0342 [+0.015, +0.055]
V4 textcat	seq-20260718-222820	0.1950	-0.0168 [-0.029, -0.006]	0.1565	+0.0335 [+0.015, +0.051]
V5 balanced-pca192	seq-20260718-222834	0.2159	+0.0042 [-0.006, +0.015]	0.1586	+0.0356 [+0.019, +0.052]
V7 std-txtcatup	seq-20260718-222848	0.2055	-0.0063 [-0.019, +0.006]	0.1488	+0.0259 [+0.006, +0.046]
V10 text-only	seq-20260718-222901	0.1999	-0.0119 [-0.025, +0.001]	0.1377	+0.0147 [-0.005, +0.034]
V6 sonic-64 (64-d)	seq-20260718-222905	0.1936	-0.0182 [-0.031, -0.005]	0.1216	-0.0014 [-0.017, +0.013]

Table 10: music@10 by predictor family, from the offline backfill over all 53 next-track runs (one shared 1431-case split). The recall@10 champion (**seq-blend**) is *not* the music@10 champion (**seq-nexttrack**). Best cell per column bolded.

Predictor family	runs	best recall@10	best music@10	mean music@10
seq-blend	23	0.223	0.461	0.441
seq-nexttrack	24	0.159	0.482	0.448
seq-markov	1	0.107	0.407	0.407
seq-ann	1	0.090	0.426	0.426
seq-stacker	3	0.087	0.418	0.411
seq-popularity	1	0.001	0.185	0.185

Interpretation. music@10 is a genuinely distinct, non-saturated view of next-track quality: it rewards *sonic* proximity, which the exact-hit metric ignores. The recall-versus-music decoupling has a clean mechanism — a model that regresses the content latent (**seq-nexttrack**) lands *near* the truth in content space even when it misses the exact next track, so its misses “sound right” more often than the blend’s, which optimizes exact rank. This is why the metric is display-only and must not silently become a selection objective: promotion still ranks by Recall@10. The metric inherits a mild circularity from the *balanced* dial (the text block embeds the artist name; the genre block is a genre prototype), so it partly rewards same-artist / genre proximity — intended for “sounds / feels alike,” not a perceptual ground truth. A companion GET /api/pathfinder/similar endpoint serves “songs like this” over the same index. *Source: 2026-07-20-musical-distance-metric.md.*

15 Session holistiness metrics and the anti-eager levers (MMR λ / eager β)

Rationale. A recurring qualitative complaint about the top blends is that their predictions feel “too album-eager / bigram-like” — they parrot the current artist rather than curating a holistic continuation. This campaign quantifies that symptom with a newly-added *display-only session-holistiness* metric suite, tests the user’s hypothesis that *the GRU beats the LSTM because it is more eager*, and measures whether two new Phase-2 levers reduce eagerness at an acceptable recall cost: an eval-time MMR re-rank (**mmr_lambda**) and an anti-eager training regularizer (**eager_beta**). It is a measurement / exploratory pass: nothing is promoted, the primary metric stays Recall@10,

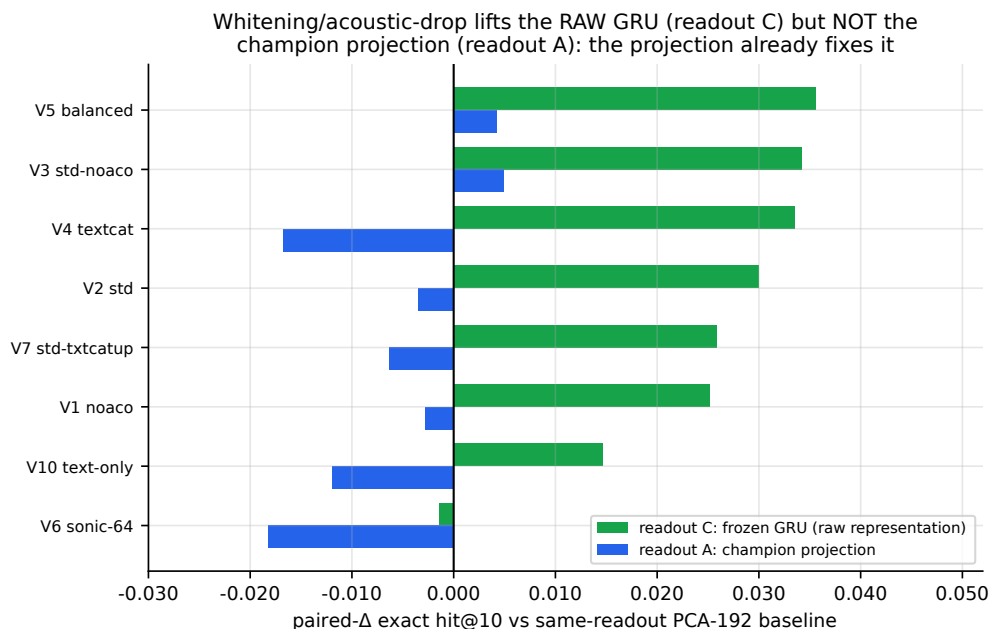


Figure 6: The representation handicap is real for the *raw* GRU but *absorbed* by the champion projection. Six of eight whitening / acoustic-drop / metric spaces lift the frozen GRU (readout C, green; all six with paired-Δ CI entirely > 0, +0.025 to +0.036), but none lifts the champion projection (readout A, blue; the two nominally-positive spaces straddle 0, and V4/V6 significantly regress). The learned InfoNCE map already re-weights away the loud acoustic / numeric directions, so whitening the input first is redundant at the champion ceiling: whitening and the projection are the same fix.

and the holistiness objective decision is explicitly deferred to the user. Every arm runs on the *canonical* PCA-192 champion split `seq-20260715-131139` (1431 test sessions; cold-item rate 0.55), and the champion reproduced its on-record Recall@10 0.2117 exactly (0.21174).

The new display-only holistiness metrics. Wired end-to-end alongside `music@10` (`seq_common.eval_from_scores` → Rust → `domain.toml` → UI); the primary metric is unchanged.

artist_adj@10 (↓). Fraction of the top-10 sharing the *seed* track’s artist — the “parrots the current artist” eagerness rate. **Lower is better.**

album_adj@10. The same on (artist, album); **omitted on this dataset** because its `items.json` carries no album field (expected, not an error — covered by a separate album-carrying dataset).

mood_coh@10 (↑). Mean cosine of the top-10 to the prefix mood centroid in the content-metric space; higher is more mood-coherent.

ild@10 (↑). Intra-list diversity (mean pairwise sonic distance) of the top-10; higher is less duplicative.

suffix_recall@10 / cont_prec@10. Multi-step continuation coverage, produced only by the on-demand harness `seq_continuation_eval.py` (Phase E below), not by normal runs.

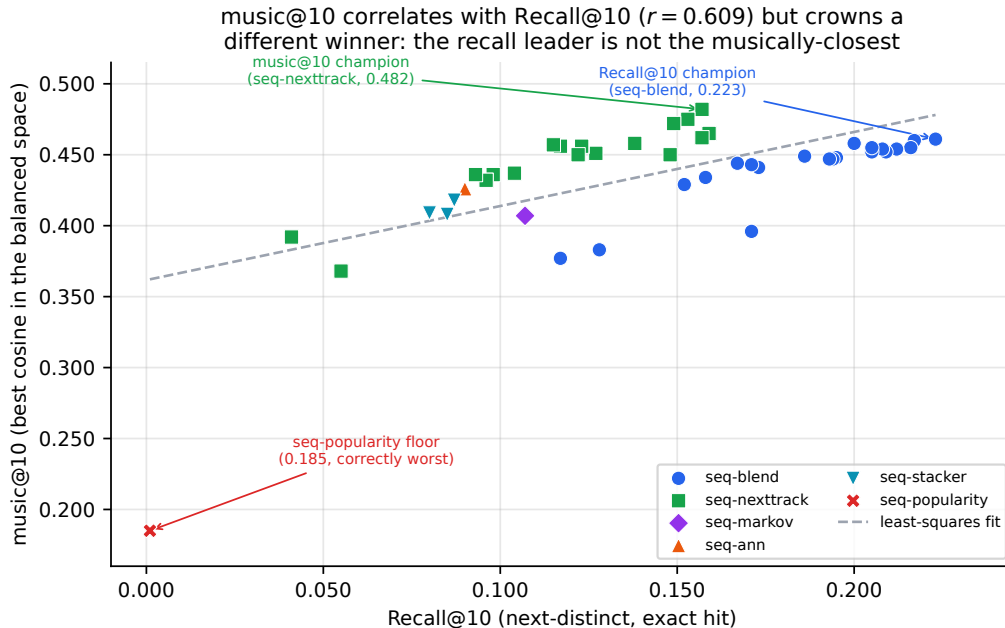


Figure 7: music@10 versus Recall@10 across all 53 next-track runs, one point per run, coloured by predictor family. The two axes correlate ($r = 0.609$: a better ranker also lands closer misses) but crown different winners — the Recall@10 champion (a **seq-blend** run at 0.223) is not the musically-closest, which is a **seq-nexttrack** run (music@10 0.482) that regresses the content latent and retrieves by cosine. The **seq-popularity** floor is correctly worst on both. music@10 is display-only; Recall@10 stays the primary metric.

The sanity gate passed: music@10 / mood_coh@10 / ild@10 populate on all blend and GRU runs, so the content-metric vectors loaded and the MMR / mood / ild levers are live, not no-ops. Both new levers are OFF by default and verified byte-identical when off.

Results — the holisticness fingerprint (Phase A). Table 11 fingerprints the model family. **Eagerness tracks Recall@10 inside the blend family** — the champion (Recall@10 0.2117) is the *most* artist-eager model measured (artist_adj@10 0.610), with R+M next (0.579) — but the eagerness lives in the *Markov* leg, not the RNN: the standalone first-order Markov bar is already at 0.543, while the single GRU / LSTM sit far lower at artist_adj@10 ≈ 0.345 . So the blends buy their Recall@10 lead by being artist-adjacent (bigram-like), which is exactly what the holisticness suite flags. The single RNNs are conversely the *mood-coherent, low-eager* models (highest mood_coh@10, lowest real-model artist_adj@10). Crucially the metrics are *not trivially gamed*: the popularity floor maxes ild@10 (0.898) and minimizes artist_adj@10 (0.003) yet scores ≈ 0 recall and *negative* mood_coh@10 — you maximize them by being random, so the meaningful low-eager / diverse models are the real single RNNs.

GRU versus LSTM (Phase B) — the hypothesis is not supported. With hyperparameters matched (h256, InfoNCE, 40 epochs, lr 1×10^{-3}), the GRU and LSTM are statistically *tied*: Recall@10 0.1230 versus 0.1188 ($\Delta +0.0042$, inside the ± 0.015 band), and artist_adj@10 0.347 versus 0.344 (a negligible $+0.003$). The GRU is not measurably more eager than the LSTM on this split, so the “GRU wins because it parrots the current artist harder” hypothesis is **refuted here**.

The MMR λ lever (Phase C) — the strong knob. MMR is a pure eval-time re-rank, so the

Table 11: Phase A holistiness fingerprint across the model family, sorted by Recall@10 (1431 test sessions). All runs share the prefix `run-20260722-165345-`; the id column lists the distinguishing suffix. Recall@10 (raw hits of 1431), MRR, artist@10, genre@10, music@10, mood_coh@10, ild@10 higher-better; **artist_adj@10 lower-better**. Champion row and best cell per metric bolded. [†] The popularity floor’s extreme artist_adj@10 / ild@10 are *degenerate* (it ignores the seed): it lands ≈ 0 recall and negative mood_coh@10, proving the metrics are not gamed low/high by a good model.

Config	id	R@10 (raw)	MRR	art@10	gen@10	mus@10	artadj@10↓	mood@10↑	ild@10↑
A1 champion $R'+M+C'$	82a8f	0.2117 (303)	0.122	0.333	0.401	0.454	0.610	0.398	0.406
A2 R+M (blend)	28dbb	0.1712 (245)	0.095	0.329	0.398	0.443	0.579	0.399	0.403
A3 GRU alone ($\beta=0$)	e39c9	0.1230 (176)	0.053	0.338	0.450	0.456	0.347	0.479	0.454
B LSTM (else matched to A3)	bb387	0.1188 (170)	0.049	0.328	0.455	0.449	0.344	0.458	0.463
A4 markov (first-order)	047cc	0.1069 (153)	0.069	0.298	0.389	0.407	0.543	0.310	0.462
A5 popularity	34614	0.0007 (1)	0.001	0.014	0.085	0.185	0.003[†]	−0.041	0.898[†]

four originally-launched per- λ training runs were *cancelled* (not failed experiments); the sweep was computed offline from the single champion checkpoint (the $\lambda=\text{off}$ cell reproduced the champion’s Recall@10 0.2117 exactly). As λ drops from off toward 0.3 (Table 12; Figure 8), Recall@10 falls — the *only* quantity traded — while ild@10 rises and artist_adj@10 falls (de-eagering). $\lambda=0.9$ is **essentially free**: the recall Δ is -0.0014 , well inside the ± 0.015 band (a tie), yet it already lifts ild@10 $0.406 \rightarrow 0.436$ and trims eagerness $0.610 \rightarrow 0.603$. $\lambda \approx 0.7$ trades ~ 2 recall points ($\Delta -0.0181$, just outside the band) for a large ild@10 gain ($0.406 \rightarrow 0.520$) and a real eagerness drop ($0.610 \rightarrow 0.571$). A notable side effect: **music@10 rises monotonically** as you diversify ($0.454 \rightarrow 0.492$) — sonic closeness and list diversity are not in tension; only exact recall is traded.

Table 12: Phase C MMR λ sweep on the champion, computed offline from its checkpoint (`mmr_pool` 200; 1431 test sessions). Recall@10 (raw hits), MRR, artist@10, music@10, mood_coh@10, ild@10 higher-better; **artist_adj@10 lower-better**. Best cell per metric bolded; the champion default ($\lambda=\text{off}$) recall / MRR bolded.

λ	R@10 (raw)	Δ vs off	MRR	art@10	mus@10	artadj↓	mood↑	ild↑
off (1.0)	0.2117 (303)	—	0.122	0.333	0.454	0.610	0.398	0.406
0.9	0.2103 (301)	−0.0014 (TIE)	0.121	0.341	0.462	0.603	0.391	0.436
0.7	0.1936 (277)	−0.0181	0.116	0.347	0.476	0.571	0.364	0.520
0.5	0.1642 (235)	−0.0475	0.108	0.359	0.487	0.486	0.315	0.653
0.3	0.1097 (157)	−0.1020	0.100	0.358	0.492	0.293	0.238	0.815

The anti-eager regularizer β (Phase D) — the weak knob. A training-time penalty on near-copy prediction of the current track’s latent (`eager_margin` 0.0) barely moves the needle (Table 13): across β 0 to 0.2, artist_adj@10 shifts only $0.347 \rightarrow 0.340$ ($\Delta -0.0074$) and ild@10 only $0.454 \rightarrow 0.458$, with Recall@10 flat (all four inside ± 0.015). The single GRU is already low-eager (0.347), so there is little copy behaviour left to penalize — β is not a useful holistiness knob at these settings.

Multi-step continuation (Phase E). The on-demand harness (`seq_continuation_eval.py -m 3 -k 10`, 1172 continuation cases) shows the GRU *beats* the Markov bigram on both multi-step metrics (Table 14): the RNN’s advantage *widens* past the immediate next track, because the bigram’s single-step co-occurrence does not compound into good 3-step continuations whereas the GRU’s learned sequence state does — the mirror image of the single-step board, where the bigram-carrying blend wins raw Recall@10.

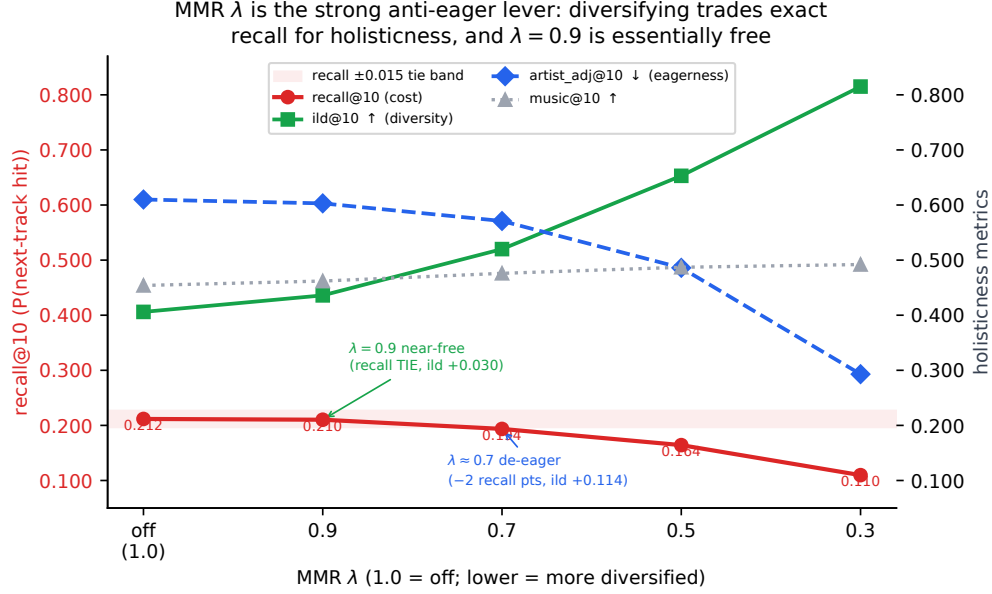


Figure 8: The MMR λ trade-off on the next-track champion. As λ drops from off toward 0.3, Recall@10 (red, left axis) falls — the only quantity traded — while ild@10 rises and artist_adj@10 falls (right axis; more diverse, less artist-eager) and music@10 even *rises*. $\lambda=0.9$ sits inside the ± 0.015 recall tie band yet already lifts ild@10 by 0.030; $\lambda \approx 0.7$ buys a large ild@10 gain and real de-eagering for ~ 2 recall points. MMR is the strong holisticness lever; β (Table 13) is weak.

Table 13: Phase D anti-eager regularizer β sweep on the single GRU (eager_margin 0.0; 1431 test sessions). Recall@10, artist@10, mood_coh@10, ild@10 higher-better; **artist_adj@10 lower-better**. Best cell per metric bolded.

run id	β	R@10 (raw)	MRR	art@10	artadj↓	mood↑	ild↑
run-20260722-165345-e39c9	0.0	0.1230 (176)	0.053	0.338	0.347	0.479	0.454
run-20260722-165345-353b5	0.05	0.1237 (177)	0.053	0.337	0.346	0.478	0.455
run-20260723-002044-af11a	0.1	0.1202 (172)	0.053	0.335	0.344	0.477	0.456
run-20260723-002044-17cf8	0.2	0.1202 (172)	0.051	0.336	0.340	0.475	0.458

Interpretation and the deferred decision. The campaign delivers a clean mechanistic account of the eagerness symptom: it is sourced from the Markov bigram leg (which parrots the current artist via raw co-occurrence), not the neural leg, so any future de-eagering should target the bigram / the fusion, not the RNN. Of the two new levers, the eval-time MMR re-rank is the *strong* knob and the training-time β regularizer is *weak*. Nothing is promoted — the primary metric stays Recall@10 and the champion is unchanged — but $\lambda \approx 0.9$ is **flagged as a near-free holisticness upgrade for a future user promotion decision** (Recall@10 tie within noise, ild@10 0.406→0.436, eagerness 0.610→0.603), with $\lambda \approx 0.7$ the “meaningful de-eagering for ~ 2 recall points” operating point; adopting either requires a user decision plus a 3-seed / native-run confirm. The β regularizer is not recommended as a lever at these settings. *Source: 2026-07-22-session-holisticness-and-antieager-levers.md.*

Table 14: Phase E multi-step continuation (harness -m 3 -k 10, 1172 cases) on seq-20260715-131139: the GRU (A3 checkpoint) beats the first-order Markov bigram on both metrics (higher-better). Winner bolded.

Model	suffix_recall@10	cont_prec@10
markov (first-order bigram)	0.0869	0.0236
GRU (gru-infonce-h256, A3)	0.0963	0.0273

16 Dual-tower fusion refuted: the fourth combiner null

Rationale. By 2026-07-25 the record held three independent *learned-combiner* nulls — the XGB stacker (0.080 to 0.087, *below* its own GRU base leg; Section 10), reciprocal-rank fusion (−0.029) and even *oracle* test-fit blend weights (+0.004, CI straddling zero; Section 11). All three were **post-hoc, score-level** combiners, which left one obvious rebuttal standing: *of course* a bolted-on combiner fails — train the fusion end-to-end so the parts co-adapt. This campaign runs exactly that rebuttal, at the *representation* level, jointly, from scratch.

The vehicle is a new predictor, **seq-dualgru**: two *independent* recurrent towers (separate weights) read the prefix in parallel over configurable *causal views*, their per-step hidden states are concatenated, and one MLP head fuses them into the predicted next-item latent. Three views are exposed, all strictly causal at step t : **latent** (the raw item vector), **delta** (the step-to-step difference, $x_{-1} = 0$), and **cummean** (the running session centroid — the quantity the **seq-ann** pooled-MLP leg was built from). Crucially it reuses **seq_nexttrack.make_batches** and **step_loss** verbatim, so objective, batching, early stopping and evaluation are byte-for-byte the family’s, and it carries no bidirectional mode (a backward pass would leak the teacher-forcing target).

The hypothesis was sharp: two towers should beat one *only if they see genuinely different views*; two towers on the same **latent** view should reduce to a capacity / implicit-ensemble effect a wider single GRU can buy more cheaply. The design therefore crosses the *complete upper triangle* of the $\{\text{latent}, \text{delta}, \text{cummean}\}^2$ view matrix (six cells at **fusion_layers** = 1) with a *fusion-depth* axis (two cells at **fusion_layers** = 0, a pure linear read of the concatenated states) and — the axis the earlier topology scan lacked (Section 5) — a **parameter-matched single-GRU capacity control**. Parameter counts were re-measured by instantiating the real classes at **latent_dim** = 192: the control C1 (h425, 871 017 parameters) matches the six **fusion_layers** = 1 dual arms (871 872) to 0.098 %, and is 10 % *larger* than the **fusion_layers** = 0 arms (789 696) — deliberately generous to the single tower.

The honest bar. Two boards were pre-registered *separately*, so that the scientific verdict and the board event could not be traded off after the fact. On **Recall@10** the bar for a standalone single model is the single-model board 0.123, *not* the three-leg blend champion 0.212 — comparing one trained model against a fusion of three legs would be a category error, and dual-tower-as-the- R' -leg is a **seq-blend** code change explicitly out of scope. On **holisticness@10** the crown rule carries an anti-degenerate recall floor (see below). All ten runs are on the canonical split seq-20260715-131139 at seed 1337, executed one at a time on a single remote worker, and the baseline C0 reproduced its on-record Recall@10 0.12299 (176 of 1431) *exactly*.

Results. Table 15 is the batch. **No arm is bolded as a winner because there is no winner**: the single-GRU baseline holds the batch’s best Recall@10, and not one of the eight dual arms beats it by paired- Δ CI above zero — six lose outright and two (the **fusion_layers** = 0

pair) tie. The pre-registered refutation rule fired.

Table 15: Dual-tower view-pairing $\times$ fusion-depth scan, ten runs on `seq-20260715-131139` (1431 test sessions), sorted by Recall@10. All arms share the run-id prefix `run-20260725-143502-`; C0 is `run-20260725-143143-97c1e`. Paired Δ is a per-session paired bootstrap against C0 (2000 resamples, rng 1337). Recall@10 raw = successes of 1431; CI bounds are given to three decimals, and a CI straddling zero is a *tie*. Arm codes carry the view pair (lat = `latent`, del = `delta`, cum = `cummean`) and the fusion depth (`-f0` = linear read, otherwise a 256-wide MLP head). R@10, MRR, mus@10, ild@10 higher-better; **artadj@10 lower-better**; H is the *3-factor* holisticness generation (Section 18). Best cell per metric bolded. [†] D-CC’s extreme artadj@10 / H are *not* a real crown: they are bought at 95 of 1431 hits, 0.0566 below the baseline, failing the ± 0.015 recall floor by $3.8\times$.

Arm	id	params	R@10 (raw)	Δ vs C0 [95% CI]	MRR	mus@10	artadj↓	ild↑	H↑
C0 h256 <i>base</i>	97c1e	394 944	0.12299 (176)	—	0.05341	0.4558	0.3474	0.4536	0.14172
C1 h425 <i>ctrl</i>	be11b	871 017	0.12089 (173)	−0.0021 [−.011, +.006]	0.05215	0.4597	0.3640	0.4506	0.13850
D-LL-f0 <i>lat/lat</i>	18b4a	789 696	0.11880 (170)	−0.0042 [−.013, +.004]	0.05631	0.4553	0.3630	0.4506	0.13774
D-LC-f0 <i>lat/cum</i>	3e5d4	789 696	0.11740 (168)	−0.0056 [−.017, +.005]	0.05087	0.4458	0.3557	0.4630	0.13648
D-LL <i>lat/lat</i>	12fe6	871 872	0.10273 (147)	−0.0203 [−.035, −.007]	0.04235	0.4446	0.3092	0.4873	0.14635
D-LC <i>lat/cum</i>	a95c5	871 872	0.09713 (139)	−0.0259 [−.039, −.013]	0.03901	0.4380	0.2997	0.4913	0.15000
D-LD <i>lat/del</i>	8d214	871 872	0.09154 (131)	−0.0315 [−.045, −.018]	0.04029	0.4385	0.2999	0.4988	0.14673
D-DC <i>del/cum</i>	e9cfb	871 872	0.08665 (124)	−0.0363 [−.050, −.022]	0.03379	0.4125	0.2319	0.5413	0.16340
D-CC <i>cum/cum</i>	88a4c	871 872	0.06639 (95)	−0.0566 [−.072, −.041]	0.02664	0.3935	0.1899[†]	0.5235	0.19715[†]
D-DD <i>del/del</i>	4324f	871 872	0.05800 (83)	−0.0650 [−.082, −.049]	0.02819	0.4016	0.2363	0.5457	0.13335

Capacity is not the missed lever either. The param-matched control C1 (h425, 871 017 parameters — $2.2\times$ C0) scores 0.12089 against C0’s 0.12299, a paired Δ of -0.0021 straddling zero. This fills a real gap: single-GRU *width* had only ever been scanned on the 64-dim AE space (Section 5), and the rich PCA-192 space was untested. It behaves the same way, so **single-GRU width is now closed on PCA-192 as well as AE-64**. Because C1 is a genuine 0.098 % parameter match to the `fusion_layers = 1` arms, the null cannot be re-attributed to “capacity would have helped”: it demonstrably would not.

View attribution — complementarity is real, but lands on the wrong pair. The complete view triangle is what makes this mechanically decidable rather than a judgment call, and the answer is genuinely surprising (Table 16, Figure 9). The pair the design nominated as its best hope — `latent + cummean` — is *not* complementary: it ties its latent/latent parent and beats only the centroid-only one, i.e. the raw-latent tower carries the cell and the centroid tower is dead weight. Same for `latent + delta`. But `delta + cummean` *is* complementary — it beats *both* parents with CI above zero. Read together: the co-adaptation mechanism the hypothesis proposed genuinely **works**; it is simply that **the raw latent view already subsumes everything the other two carry**. A GRU over raw latents integrates step-to-step movement and the running mean internally, so pairing it with an explicit computation of either is redundant — while two *impoverished* views do have non-overlapping information to pool, at a hopeless absolute level (0.08665). That is a clean mechanistic explanation for the null, not merely an absence of effect.

The MLP fusion head is actively harmful, not merely useless. Both `fusion_layers = 0` arms beat their `fusion_layers = 1` twins by paired- Δ CI above zero (+0.0161 on latent/latent, +0.0203 on latent/cummean) at 10 % *fewer* parameters — so removing the head makes the model measurably better in *both* view regimes. Mechanistically this fits the whole family arc: the objective is cosine/InfoNCE retrieval *in* the item-latent space, so the head’s job is to land *in* that geometry, and an extra 256-wide non-linear layer between the recurrent state and a metric read-out just adds a warp the loss must then undo. The trade-off runs the other way on holisticness (both f0 arms have *lower* H, Δ H CI below zero): the f1 arms’ extra warp diffuses the top-ten (higher ild,

Table 16: View attribution: each mixed cell against *both* of its single-view parents (paired Δ Recall@10, 1431 sessions). Only one of three mixed pairs is genuinely complementary — and it is the one with no raw-latent tower.

Mixed cell	vs parent 1	vs parent 2	Verdict
D-LC <i>lat/cum</i>	D-LL: -0.0056 [$-.018, +.007$]	D-CC: $+0.0308$ [$+.018, +.045$]	<i>not</i> complementary
D-LD <i>lat/del</i>	D-LL: -0.0112 [$-.023, +.001$]	D-DD: $+0.0335$ [$+.020, +.047$]	<i>not</i> complementary
D-DC <i>del/cum</i>	D-DD: $+0.0287$ [$+.015, +.043$]	D-CC: $+0.0203$ [$+.008, +.034$]	complementary: both CI>0

lower artist_adj) at the cost of exact hits. The rule adopted is **default fusion_layers** = 0, later confirmed a third time (Table 19).

Holisticness and recall are anti-correlated, and the recall floor earned its place. Five arms lift holisticness@10 by paired- Δ H CI above zero, topping out at D-CC’s 0.19715 (+0.055) — which on the naked crown metric would have ranked second on the entire board. But D-CC lands 95 of 1431 exact hits against the baseline’s 176. The ordering of the batch by H is almost the *reverse* of its ordering by recall (Figure 10), so **all five Δ H winners fail the pre-registered ± 0.015 recall floor** and the crown rule never fired. Without that floor the campaign would have “won the crown” with a model 46 % worse at its actual job — precisely the degenerate **mood-session** failure mode (H 0.2332 at Recall@10 0.0238) the floor was written to block. On the present evidence, **unconstrained holisticness is maximized by predicting less well**, so any future holisticness objective on this corpus must be a *constrained* one.

Interpretation. Nothing was registered and nothing promoted; the champion, the leaderboard and the pins are untouched. What the campaign buys is a *closure*: the combiner null is now robust across the whole combiner design space on this corpus — post-hoc score-level (three ways) *and* representation-level jointly-trained with a parameter-matched control — which materially consolidates “the fixed equal-thirds z-blend is the combiner of record.” Two genuine sub-findings survive the null (the harmful MLP fusion head, and **delta+cummean** complementarity), and the machinery itself is sound: 10 of 10 runs succeeded, the new predictor trains, converges and evaluates at exact parity with the family, and D-LL-f0 even posts the batch’s best MRR (0.05631 against C0’s 0.05341) — the one metric on which a dual arm leads. Per the noise-band discipline that is “at least equal, likely neither better nor worse,” not a win.

A provenance finding that revises two earlier numbers. Before the batch, the live remote worker directory was found *stale*: its `seq_common.py` had no `holisticness_at_k` at all, `seq_nexttrack.py` differed from the hub by md5, and three `seq_*.py` modules were missing entirely. After refreshing and verifying the whole family md5-identical to the hub, C0 returned 0.12299, bit-identical to the hub’s on-record value — *not* the 0.1201957 that two earlier remote runs (run-20260723-002044-af11a, -17cf8) had returned. The “cross-host reduction-order offset of -0.0028 ” those runs had suggested is therefore **withdrawn: it was a code difference, not a host effect**. Their recall numbers belong to a superseded code state and are not code-comparable to later batches; they remain valid as the β -eagerness measurements they were taken for (Table 13), since those conclusions rest on within-batch comparisons. The general lesson — a divergent remote checkout produces a *wrong number*, not a crash — is now a standing pre-flight gate. *Source: 2026-07-25-nexttrack-dual-tower-fusion-scan.md*.

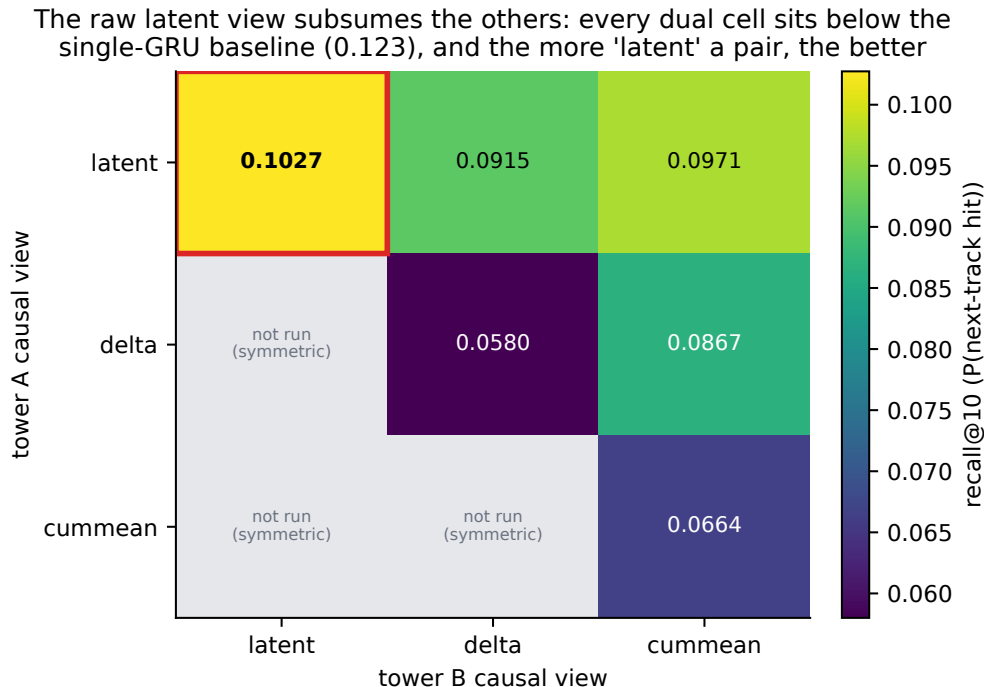


Figure 9: The complete view-pairing matrix at `fusion_layers = 1`, one run per cell. Every dual cell sits *below* the single-GRU baseline (0.12299), and the ordering is monotone in how much raw **latent** the pair contains: both-latent is best, both-delta worst. The lower triangle was never run (the pairing is symmetric by construction) and is masked in grey rather than mirrored, so the colour scale reflects only measured cells. The gradient is the finding: a recurrent tower over raw latents already computes what **delta** and **cummean** make explicit.

17 The per-step pre-encoder refuted, and the rectifier mechanism

Rationale. The dual-tower null closed *deterministic* view asymmetry: **delta** and **cummean** are lossy functions of what a raw-latent tower already holds, so pairing them buys nothing. This campaign asks the strictly stronger version — does a **learned** per-step re-embedding, which is *not* a function of information the tower already has, enlarge the hypothesis space usefully? The axis is a nonlinear re-embedding in front of the recurrence, $[\text{Linear}(d_{\text{in}}, w), \text{ReLU}, \text{Dropout}] \times \text{pre_layers}$, applied per timestep (strictly causal), with the GRU’s `input_size` becoming w .

The hypothesis has an unusually sharp analytic edge. A GRU’s own input transform is *already linear* ($W_i x$ per gate), and a *linear* pre-encoder at $w \geq d_{\text{in}}$ is exactly as expressive as a bare GRU, since $W_i(Vx) = (W_i V)x$ with no rank constraint. **The ReLU is therefore the only expressivity change**, which means no linear-pre control run is needed and a null cannot be explained away as “the pre-encoder was redundant by construction.” The design’s decisive move is the **single-tower ablation S1** — a $\text{MLP}_{256} \rightarrow \text{GRU}_{256}$ against a parameter-matched plain GRU — because without it any dual-tower result is uninterpretable: pre-encoder, or second tower?

Ten runs, same canonical split, same single remote worker, zero failures. Two reproduction gates both passed *bit-identically*: C0 returned Recall@10 0.12299091544374564 (176 of 1431, identical to the last digit, a seventh reproduction), confirming the `pre_hidden` edit did not perturb the family’s default code path; and C3, the bare dual-tower arm, returned 0.10272536687631029, exactly its on-record value, validating every dual verdict.

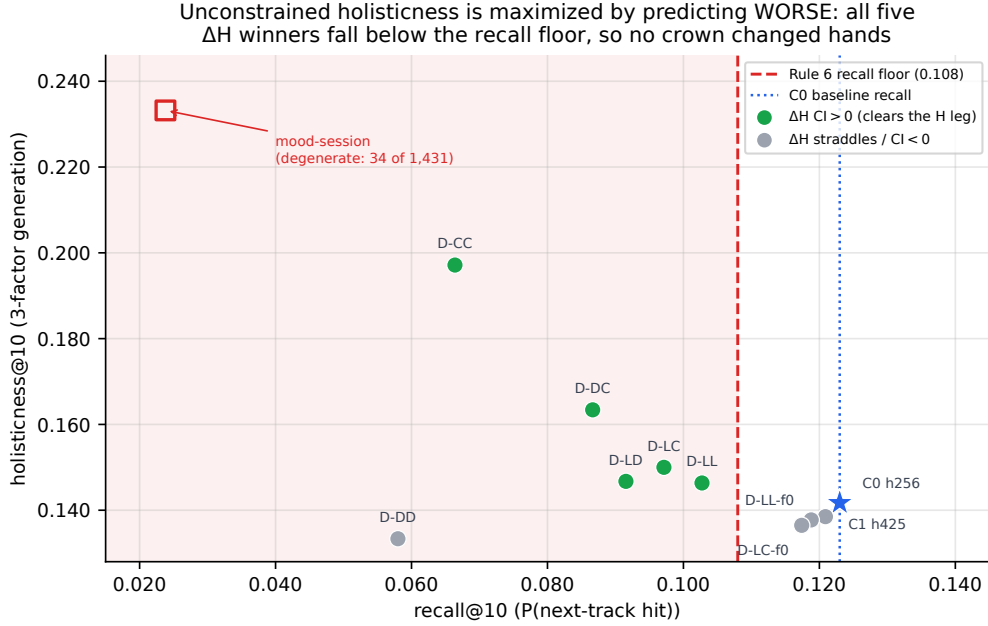


Figure 10: Why the crown rule needs a hard recall floor. Across the ten arms, holistiness@10 runs almost exactly *backwards* along Recall@10: the five arms that clear the ΔH CI-above-zero leg (green) are the five furthest *below* the baseline on exact recall, and all five fall outside the ± 0.015 floor (shaded). The open square is the incumbent **mood-session** model that motivated the floor — the highest holistiness on record at 34 of 1431 correct next tracks. H values are the 3-factor generation (Section 18).

Results. Table 17 is the batch and Figure 11 is the verdict at a glance. Not one of the six pre-MLP arms beats the baseline by paired- Δ CI above zero: five lose outright, one ties. **And the decisive ablation does not merely tie — it loses:** S1 at 0.10901 against C0’s 0.12299, $\Delta -0.0140$ with the CI entirely below zero, 156 hits against 176. Since a linear pre-encoder here is provably expressivity-neutral, **the ReLU re-embedding is destroying prefix information that the GRU’s own gated linear input transform preserved.**

The dose curve is flat, and placement is irrelevant. Sweeping `pre_hidden_b` through $\{0, 128, 256, 384\}$ at `fusion_layers = 1` (Table 18), every adjacent pair straddles zero and so do the endpoints — a **flat** axis, which by the pre-registered clause would have blocked registration even had a win rule fired. But note the *nominal* ordering is downward-then-flat, never upward: every dosed arm sits below the bare dual. Taken with S1’s significant loss, the sign of this axis is consistently negative across both towers and all four widths, so there is no interior peak to chase. Symmetry is likewise a non-factor: T1 (asymmetric, one bare tower) against T4 (both towers rectified) straddles zero on recall, so keeping tower A bare is not load-bearing at this dose.

The one solid positive: the fusion-depth regression, measured a third time and bigger. T1-f0 beats T1 by $+0.0266$ with the CI above zero, against the $+0.016$ and $+0.020$ measured in the two dual-tower view regimes (Table 19). The mechanism holds and sharpens: every extra nonlinear layer between the recurrent state and the metric read-out adds a warp the InfoNCE loss must undo, and *the more nonlinearity already added on the input side, the more the output-side warp costs.* Note too that T1-f0 — the only pre arm that ties the baseline — is exactly the arm with *both* a bare tower A *and* `fusion_layers = 0`: the shortest un-warped path from raw latents to

Table 17: Pre-encoder presence $\times$ placement $\times$ dose, ten runs on seq-20260715-131139 (1431 test sessions), sorted by Recall@10. All arms share the prefix run-20260726-161801-; C0 is run-20260726-161429-bdcbc. Paired Δ is a per-session paired bootstrap against C0 (2000 resamples, rng 1337); CI bounds are given to three decimals, and a CI straddling zero is a *tie*. Dual-arm codes give the two towers’ pre-encoder widths (A0/B256 = tower A bare, tower B rectified at width 256) and the fusion depth. H here is the *4-factor* (grounded) holisticness generation and is **not** comparable to Table 15’s 3-factor column (Section 18). Best cell per metric bolded; **artadj@10 lower-better**. [†] T4’s minimal artadj and T1’s maximal ild are the familiar de-eagering-by-predicting-worse trade: both sit ~ 0.025 below C0 on recall.

Arm	id	params	R@10 (raw)	Δ vs C0 [95% CI]	MRR	mus@10	artadj↓	ild↑	H↑
C2 h454 <i>ctrl</i>	7646d	969 936	0.12509 (179)	+0.0021 [−.007, +.011]	0.05389	0.4572	0.3668	0.4466	0.045765
T1-f0 A0/B256, f0	a6423	888 256	0.12369 (177)	+0.0007 [−.012, +.013]	0.05470	0.4575	0.3627	0.4616	0.044541
C0 h256 <i>base</i>	bdcbc	394 944	0.12299 (176)	—	0.05341	0.4558	0.3474	0.4536	0.047086
C1 h297 <i>ctrl</i>	7bab3	494 697	0.11321 (162)	−0.0098 [−.017, −.003]	0.05199	0.4513	0.3602	0.4474	0.045285
S1 preMLP256	ad53f	493 504	0.10901 (156)	−0.0140 [−.027, −.001]	0.04463	0.4499	0.3240	0.4815	0.047132
C3 bare <i>dual</i> , f1	0df6f	871 872	0.10273 (147)	−0.0203 [−.035, −.007]	0.04235	0.4446	0.3092	0.4873	0.046612
T2 A0/B128, f1	dc7d2	847 424	0.09993 (143)	−0.0231 [−.037, −.009]	0.04490	0.4415	0.3039	0.4939	0.046431
T3 A0/B384, f1	ca0b8	1 093 440	0.09993 (143)	−0.0231 [−.037, −.009]	0.04333	0.4441	0.3068	0.4961	0.045895
T4 A256/B256, f1	bd5b5	1 068 992	0.09783 (140)	−0.0252 [−.039, −.011]	0.03829	0.4343	0.2845 [†]	0.4986	0.043985
T1 A0/B256, f1	9450b	970 432	0.09713 (139)	−0.0259 [−.040, −.013]	0.04176	0.4369	0.2962	0.5016 [†]	0.045769

Table 18: The dose curve, as measured: paired Δ Recall@10 against the bare dual-tower arm C3 at fusion_layers = 1. Flat by the pre-registered test — but nominally downward, never upward.

pre_hidden_b	Arm	R@10 (raw)	Δ vs C3 (bare)
0	C3	0.10273 (147)	—
128	T2	0.09993 (143)	−0.0028 [−0.0147, +0.0091] straddles
256	T1	0.09713 (139)	−0.0056 [−0.0189, +0.0077] straddles
384	T3	0.09993 (143)	−0.0028 [−0.0154, +0.0105] straddles

the read-out. fusion_layers = 0 is the family default, full stop.

The mechanism, measured rather than inferred. The analytic argument named the rectifier as the only candidate; a follow-up interpretability read then *measured* it (Figure 12). The decisive control was a counterfactual tap the sparse-autoencoder engine cannot express — the model’s *own* affine map with the ReLU deleted. Reading next-item genre decodability at the same step (11 184 probe test rows, 25 classes, paired standard error ≤ 0.0065): the raw latent scores 0.4033, the affine map 0.4084 (**neutral**, $\sim 0.8\sigma$, exactly as $W_i(Vx) = (W_iV)x$ predicts), and the rectified tap 0.3700 ($\sim 5.9\sigma$ — **the ReLU is the whole loss**). The smoking gun is arithmetic: frac_exact_zero on the rectified tap is 0.5327, matching the affine tap’s frac_negative_coords 0.5327 to four digits. The rectifier deletes *precisely* the negative half, leaving ~ 120 live units for 192 signed directions — *fewer live coordinates than the input has dimensions*. PCA-192 latents are zero-centred and near-symmetric by construction, and at $w = 256$ (only $1.33\times$ the input dimension) there is nowhere near enough over-completeness to re-encode the discarded half-space, unlike the wide embeddings where ReLU MLPs normally live. The damage survives the recurrence: within one model, tower_b (fed rectified) scores 0.3929 against its *identical twin* tower_a (fed raw) at 0.4183, $\sim 3.9\sigma$ — and tower_a reproduces the standalone plain GRU (0.4235), which is exactly what “the bare tower rescues” requires.

Interpretation, and one honest caveat about the controls. This is the fifth consecutive architecture null, and it closes the *input side* as the view null closed the input-view direction: together

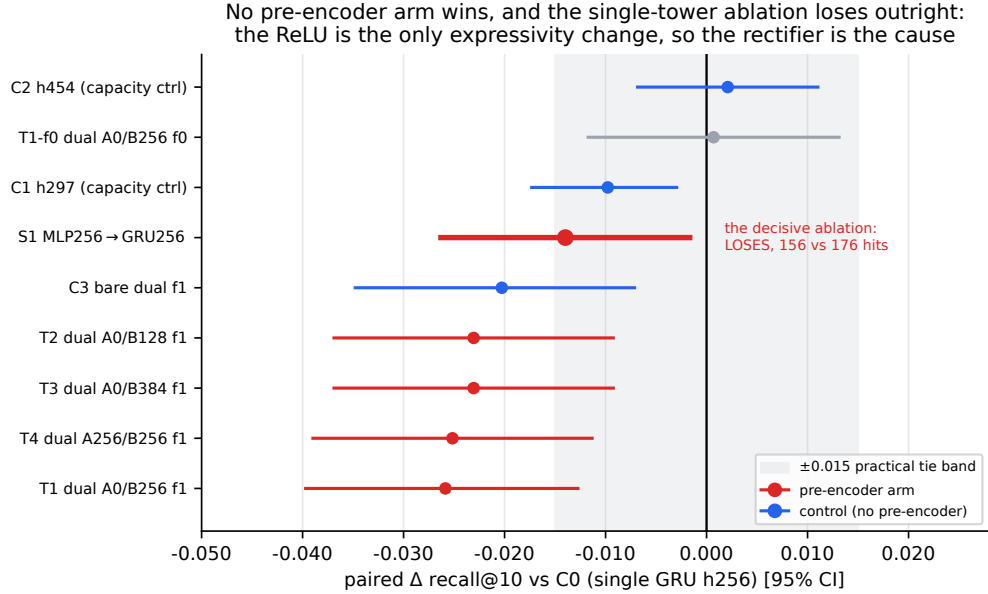


Figure 11: The pre-encoder verdict. Paired per-session bootstrap Δ Recall@10 against the reproduced baseline, one row per arm; the shaded strip is the ± 0.015 practical tie band. Not one pre-encoder arm reaches a CI entirely above zero. The decisive row is S1 — the single tower, the only arm whose result is unambiguously attributable to the pre-encoder — which loses outright. Because a *linear* pre-encoder at this width is provably expressivity-neutral, the rectifier is the only candidate cause.

with the four learned-combiner nulls, adding representation machinery *around* a single GRU over a good latent space does not help on this corpus, whereas every real gain on record came from improving the *latent space itself* (the InfoNCE content projection, PCA-192) or from the fixed z-blend. The axis is now quantitatively bounded — best pre arm 0.12369, a tie; dose flat to twice the input dimension; placement irrelevant; the fusion-depth interaction resolved. The caveat is that the *parameter-matched width control was a weaker instrument than assumed*: C1 (h297) came in at 0.11321, itself losing to C0, and the batch’s width points run 176 → 162 → 173 → 179 hits at h256 / h297 / h425 / h454 — non-monotone, spanning 0.0119 in recall with no trend. “Single-GRU width is flat on PCA-192” survives as a *trend* claim (h454 ties h256), but per-width jitter is $\sim \pm 0.01$ on a single split, roughly two-thirds of the noise band. **A width-matched control that is itself a single unreplicated run is not a reliable capacity anchor at this effect size** — which is why it matters that S1’s verdict rests on the C0 comparison, reproduced to the last digit. A second caveat is un-disentangled: the pre-MLP ends in Dropout, so pre arms receive GRU-*input* dropout that the bare arms lack, and S1’s loss could be partly over-regularization rather than the rectifier — separating them needs a `pre_dropout` parameter that does not yet exist. *Source: 2026-07-26-nexttrack-pre-encoder-scan.md.*

18 Reading the record: statistical power and the holistichness generations

Two cross-cutting facts emerged from the 2026-07-25/26 batches that change how every number above should be read. Neither is a result about a model; both are results about the *measuring*

Table 19: The `fusion_layers` regression, three independent measurements. Paired Δ Recall@10 of the linear-read arm minus its MLP-head twin; positive means removing the fusion MLP *helps*. The ΔH column is available only for the two same-batch 3-factor pairs. *Sources: 2026-07-25-nexttrack-dual-tower-fusion-scan.md (rows 1–2), 2026-07-26-nexttrack-pre-encoder-scan.md (row 3).*

Pair (f0 – f1)	Regime	Δ R@10 [95% CI]	ΔH [95% CI]
D-LL-f0 – D-LL	redundant views, no pre-encoder	+0.0161 [+0.0021, +0.0301]	–0.0086 [–0.0116, –0.0055]
D-LC-f0 – D-LC	complementary views, no pre-encoder	+0.0203 [+0.0063, +0.0335]	–0.0135 [–0.0171, –0.0100]
T1-f0 – T1	with a per-step pre-encoder	+0.0266 [+0.0140, +0.0391]	—

instrument, and they belong here rather than buried in one campaign.

18.1 The nulls are power-limited, and that is a different claim

Every verdict in this document rests on a paired per-session bootstrap over the same 1431 test sessions. That n has a measurable resolution. Averaged over the nine paired comparisons of the pre-encoder batch the 95% CI half-width is 0.0124 on Recall@10 (0.0128 over the dual-tower batch’s nine, so the figure is stable), and ~ 0.0018 on *holisticness*@10. Now compare those to the *entire spread* of an architecture family: 0.028 on Recall@10 and 0.0031 on *holisticness*@10 (Table 20). The ratio is the whole story — a family’s full range is only ~ 2.3 resolution widths wide on recall and ~ 1.9 on *holisticness*, so of eight adjacent arms roughly **two tiers** are separable on Recall@10 and **one** on *holisticness*@10. Resolving a 0.005 Recall@10 difference at this precision would require $n \approx 1431 \cdot (0.0124/0.005)^2 \approx 8800$ test sessions, six times what the split provides (Figure 13).

Table 20: Measured resolution of the test split. Half-widths are means over the nine paired-bootstrap comparisons of the pre-encoder batch (2000 resamples, $n = 1431$); the family spread is that batch’s full range on each metric. “Tiers” is spread/half-width, i.e. how many genuinely distinguishable levels eight adjacent arms can occupy. *Sources: 2026-07-25-nexttrack-dual-tower-fusion-scan.md, 2026-07-26-nexttrack-pre-encoder-scan.md.*

Metric	CI half-width at $n = 1431$	family spread	tiers	n for $\Delta = 0.005$
Recall@10	0.0124	0.0280	~ 2 of 8	~ 8800
<i>holisticness</i> @10	0.0018	0.0031	~ 1 of 8	—

The honest reading of the five architecture nulls follows directly, and it is not the reading a casual glance gives. What these campaigns established firmly is that the tested architectures do not deliver *large* gains, and that several of them do real *harm* — the harms are the well-powered results, because they are big (S1 -0.0140 ; the `fusion_layers` regression $+0.0266$; six of eight dual arms below the baseline). What the campaigns *cannot* establish is the absence of a moderate gain: a genuine $+0.005$ improvement — which on a 0.123 baseline would be a respectable 4% relative lift — would have been invisible by construction. So “dual-tower fusion is refuted” and “the pre-encoder is refuted” should be read as *no large effect, and no effect of the hypothesized magnitude*, not as “these architectures do not work.” The practical consequence is a prioritization rule rather than a caveat: on this split, further architecture arms are a poor use of runs no matter how they come out, because the design space’s whole dynamic range is barely twice the measurement error. Either enlarge n or move to axes whose effects are large — which is, empirically, exactly what the representation and supervision axes have delivered (the content

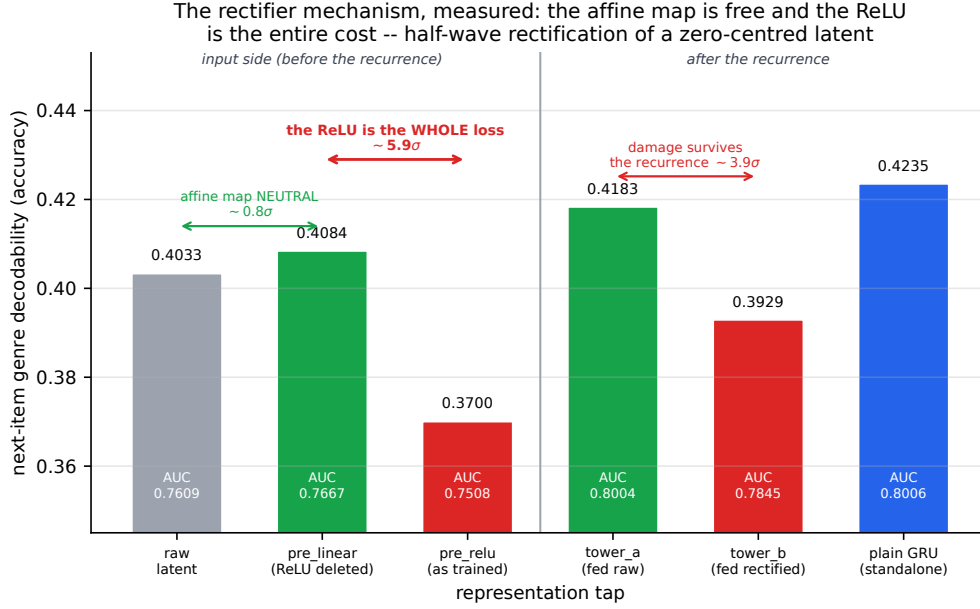


Figure 12: The rectifier mechanism, measured. Next-item genre decodability at each representation tap, including one counterfactual the interpretability engine cannot express: the model’s own affine map with the ReLU deleted. On the input side the affine map is *free* ($\sim 0.8\sigma$) and the ReLU is the *entire* cost ($\sim 5.9\sigma$); after the recurrence the damage persists ($\sim 3.9\sigma$ between two identical towers that differ only in whether their input was rectified), while the raw-fed tower reproduces the standalone plain GRU. `frac_exact_zero` on the rectified tap (0.5327) matches the affine tap’s `frac_negative_coords` (0.5327) to four digits: half-wave rectification of a zero-centred latent deletes exactly the negative half, leaving ~ 120 live units for 192 signed directions.

projection’s $+0.0259$ is twice the half-width, and the second-split hardening’s $+0.0426$ is three times it).

18.2 Three generations of `holisticness@10`, and why the board is denominated in recall

The composite *holisticness@10* was revised twice in four days, and stored values are **not comparable across the cuts**:

Gen. 1 — 3-factor (to 2026-07-24). The bare product

$$\text{clamp}(\text{mood_coh}, 0) \times \text{ild} \times (1 - \text{artist_adj}),$$

carrying *no correctness term at all*. Values land in 0.063 to 0.233 — the column in Table 15.

Gen. 2 — 4-factor, grounded (2026-07-25). Multiplies in a `music@10` grounding factor,

$$\text{clamp}((\text{music@10} - 0.185)/0.815, 0, 1).$$

Because `music@10` falls in lockstep with recall, this cancels the de-eagering “wins” the 3-factor form rewarded; values land in 0.044 to 0.050 — the column in Table 17.

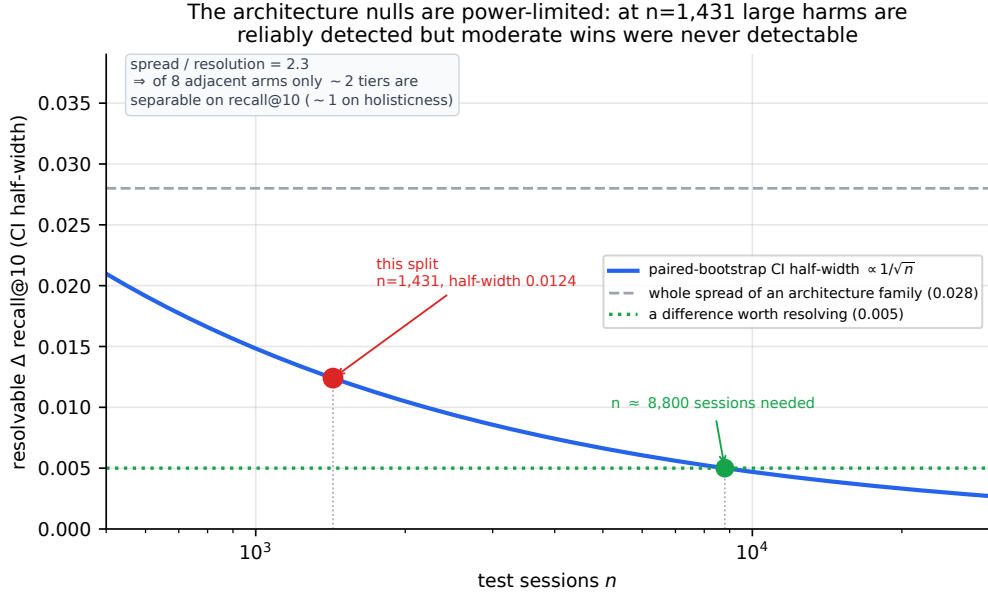


Figure 13: The architecture nulls are power-limited. The paired-bootstrap CI half-width falls as $1/\sqrt{n}$; at the split’s $n = 1431$ it is 0.0124 on Recall@10, while a whole architecture family spans only 0.028. Resolving a 0.005 difference would need ~ 8800 test sessions. The honest reading of the five architecture nulls follows directly: *large harms were detected reliably, moderate wins were never detectable at all.*

Gen. 3 — concentration-aware (2026-07-26). Eagerness becomes `max(artist_adj, artist_conc)` and the grounding floor rises $0.185 \rightarrow 0.34$. Values land near 0.009 to 0.025.

Every facet is stored per row, so any generation is recomputable after the fact (the helper `rescore_holisticness.py` does this for the runs it can reach), and Table 21 re-derives all three for a representative slice. Two consequences matter for reading this document. First, **a holisticness column must always be labelled with its generation**, because the generation effect dwarfs the model effect: every 3-factor value mechanically outranks every 4-factor value, since the grounding factor multiplies by ~ 0.3 . Second, and more importantly, **the recall and holisticness boards point in opposite directions at every generation**. The champion blend — 0.21174 Recall@10, 303 of 1431 — ranks *last* in Table 21 under all three formulas, because it is the most artist-eager model on record; the degenerate `cummean/cummean` arm and `mood-session` lead. Generation 3 demoted `mood-session` from first to a genuine zero, but it did *not* fix the dual-tower degenerate arm, so the ± 0.015 recall floor remains mandatory on any holisticness claim.

What is primary, and what this document sorts on. For completeness: the `[metrics].primary` key in `domain.toml`, and the best-models group’s `primary_metric`, have both read `holisticness@10` since 2026-07-23, so the *server-effective* primary metric is no longer Recall@10. Statements in the earlier sections that “the primary metric stays Recall@10” were true when written and are superseded *for the server*. This document nevertheless continues to denominate its leaderboard (Table 1) and every campaign verdict in **Recall@10**, deliberately: it is the record’s comparability currency, it is stable across all three metric generations, and — as Table 21 shows — a holisticness ordering would place

Table 21: The same models under all three holisticness generations, sorted by Recall@10. The generation effect dwarfs the model effect, and the ordering is roughly *inverse* to Recall@10 throughout — which is why this document’s leaderboard stays denominated in Recall@10. Dashes are runs whose facets could not be re-derived under that generation (remote-only runs with no materialized predictions file). *Sources: 2026-07-25-nexttrack-dual-tower-fusion-scan.md, 2026-07-26-nexttrack-pre-encoder-scan.md.*

Model (run)	R@10 (raw)	gen-1 (3-f)	gen-2 (4-f)	gen-3 (current)
<i>R'</i> + <i>M</i> + <i>C'</i> blend (1c1a3)	0.21174 (303)	0.0632	0.020830	0.00860
single GRU h454 (7646d)	0.12509 (179)	—	0.045765	0.02399
single GRU h256 (97c1e, bdcbc)	0.12299 (176)	0.14172	0.047086	0.02454
single GRU h425 (be11b)	0.12089 (173)	0.13850	0.046677	0.02487
S1 pre-MLP→GRU (ad53f)	0.10901 (156)	0.14499	0.047132	0.02410
D-CC <i>cum/cum</i> (88a4c)	0.06639 (95)	0.19715	0.050436	—
mood-session (degen.)	0.02380 (34)	0.23319	0.043881	0.00000

the servable champion last. Holisticness is reported throughout as a labelled, generation-tagged diagnostic column. *Sources: 2026-07-25-nexttrack-dual-tower-fusion-scan.md, 2026-07-26-nexttrack-pre-encoder-scan.md.*

19 The walk surface: measuring what the product actually serves

Rationale. Every verdict up to this point — including the crown board of Table 1 and all three holisticness generations of Section 18.2 — is computed on a **one-shot top-10**: the model sees one held-out prefix, ranks the catalog once, and is scored against one target. But the artifact this lab ships, `clients/infinite-playlist`, serves an **autoregressive walk**: it picks a track, appends it to the session, re-scores, and repeats for as many steps as the listener asks for. Those are different surfaces, and `domain.toml` has said so in prose the whole time — its `[metrics]` block states the goal as “mood-coherent, non-eager *sessions*, not guessing the literal next track.” The stated intent and the implemented metric had diverged. This section closes that gap: it defines four walk-level metrics calibrated against the user’s own listening, adds three *retrieval-time* controls, and runs a matched-sample head-to-head. It is also the one place in this document where a documented refutation is *reversed* on a new objective, which is why it is reported at length rather than folded into the campaign sections above.

19.1 The surface mismatch, and why drift was invisible

Two mechanical facts explain why nothing above could have caught a drifting journey.

First, the anchor grows. The `mood_coh` facet of `holisticness@10` measures each candidate’s cosine to the *prefix* centroid (`seq_common.py:558`). On a one-shot evaluation the prefix is fixed and that is a reasonable read of “does this list stay on vibe.” On a walk the prefix *is* the journey so far, so the reference point travels with the model: a journey that leaves its starting region by a thousand individually legal steps scores well *at every step*, because each step is compared to a centroid that has already moved most of the way there. **Drift is invisible by construction** — not under-weighted, but unmeasurable in this formulation.

Second, the composite is dominated by relevance. Differentiating the generation-3 composite at the champion’s operating point gives $\partial H / \partial \text{music@10} = +0.206$ against $\partial H / \partial \text{eager} = -0.038$: `holisticness@10` is $5.5\times$ more sensitive to its relevance factor than to its eagerness factor.

So even the term that was *meant* to price session quality is mostly re-reporting graded relevance. This is the same asymmetry that Section 18.2 observed from the outside, where the grounding factor’s ~ 0.3 multiplier made every generation-2 value rank below every generation-1 value; here it is quantified at the derivative.

19.2 Four walk-level metrics, calibrated against the user’s own sessions

The new instrument (`tools/walk_eval.py`) generates journeys from held-out *real* sessions — a 5-track seed taken from the session, then 20 autoregressive steps, prefix items excluded, at most one track per artist unless stated — and scores them on four axes:

1. **Transition match (primary).** Do the walk’s step-to-step distances look like this user’s *real* listening? This follows the playlist-lab finding directly: measure *transitions*, not averages, and calibrate against the user’s own sessions. Averaging a distance-to-centroid is the wrong shape — it penalizes a legitimate genre move and rewards standing still. Reported as the 1-Wasserstein distance (W_1 , lower better) between the journey’s consecutive-pair cosine distribution and the real one, plus two legible reads: the median step cosine and the share of steps inside the real p10–p90 band.
2. **Vibe and drift.** The *seed-anchored* cosine: mean cosine of the generated tracks to the fixed seed-core centroid (**vibe**, higher better), and the end-quarter minus start-quarter difference (**drift**, nearer zero better). The anchor is fixed at the seed and does *not* grow, which is precisely the defect of `mood_coh` that Section 19.1 identifies.
3. **Variety as credit.** Distinct artists and distinct genres in the journey, counted as a *credit* rather than as the cost that `artist_adj/artist_conc` price. Cohesion is not one genre: different genres are welcome if they preserve the vibe, so a journey covering more of them at equal vibe is better, not worse.
4. **Grounding.** Graded relevance (best sonic cosine, in the balanced musical-distance space of Section 14) between each generated track and what the user *actually* played next in that session. This is the guard-rail: without it a pleasant unrelated drift wins, which is exactly the lesson the crown learned when `music_rel` was added.

The calibration, and the single number that reframed the project. The reference distribution is the user’s own consecutive-pair cosines over the 5723 *train* sessions of seq-20260715-131139: **73 632 pairs, median 0.261, p10–p90 [−0.192, 0.830], and 27.0% of real steps *negative*.** Real listening moves — it changes direction more than a quarter of the time. Against that, the engine the showcase actually shipped (a single GRU as `gru.onnx` with a seed anchor of 0.4) walks with a median step of 0.661, roughly $2.5\times$ tighter than the user does (Figure 14). The complaint that the showcase felt “flat despite lots of artists” now has a measurement behind it: many small, safe moves. The blend champion has the opposite failure — its median step, 0.317, sits essentially on the real median, but its **vibe** is only 0.187, because its InfoNCE projection (Section 11) encodes “what follows this,” not “what sounds like this,” so a mood anchor applied *in* that projected space caps at 0.44 while the same anchor in the raw PCA-192 space reaches 0.58. Table 22 is the full first read.

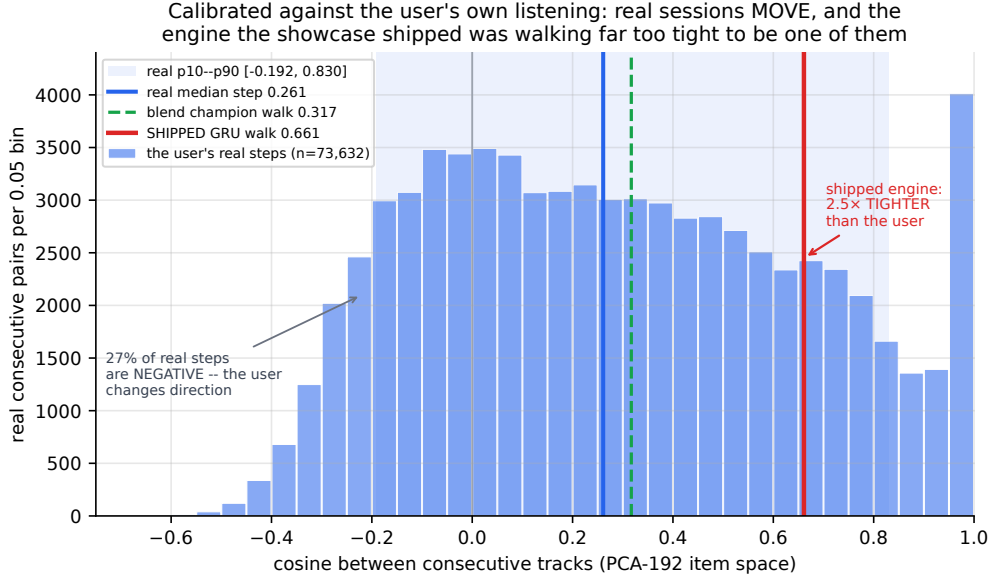


Figure 14: The core argument of this section, in one figure. The user’s own 73 632 consecutive-transition cosines (PCA-192 item space, train sessions only) are a *broad* distribution: median 0.261, p10–p90 $[-0.192, 0.830]$, 27.0% of steps negative. The engine the showcase shipped walks at a median step of 0.661 — about 2.5 $\times$ tighter than the listener it is imitating — and no one-shot metric could see it, because *holisticness@10* scores a top-10 and its *mood_coh* anchors to a prefix centroid that grows with the walk. The blend champion (0.317) matches the real stride but cannot hold a vibe. Bin counts transcribed from the calibration block of `tools/walk_eval.py`.

W_1 is not pairable, and that mattered. The distributional W_1 is computed over a *pooled* sample of steps, so it has no per-session value and cannot enter a paired bootstrap — the statistic every other verdict in this document rests on. The first two sweeps therefore compared arms measured on *different* samples (20 sessions $\times$ 20 steps for the frontier, 12×15 for the dual-tower probe), and a vibe gap of 0.588 against 0.624 is well inside what that mismatch alone can produce: nothing was established. The head-to-head (Section 19.4) therefore replaces W_1 with **stride_err** = $|\text{median step} - 0.261|$, a per-session quantity that pairs, at the cost of describing only the distribution’s centre.

19.3 Three retrieval-time controls, and two refuted ones

None of the three controls below touches training. All are applied at *retrieval* time, on the scores the model already produces, which is what makes them cheap to ship and cheap to re-tune.

ANCHOR (*a*). Add $a \cdot \cos(\text{candidate}, \text{seed centroid})$ to every candidate’s score. It holds the geist by pulling the walk back toward the region the listener seeded it from. For the champion the term is z-normalized over the admissible candidates, because its base score is itself a z-blend of three legs and adding a raw cosine to a z-blend is scale-mismatched.

STRIDE (*s*). Subtract $s \cdot \cos(\text{candidate}, \text{previous track})$. It *enlarges* each step without letting the walk leave the seed’s neighbourhood, because the anchor still applies — the two terms are orthogonal by design, one pulling toward a fixed point and one pushing off the last point. It is deliberately *deterministic*: sampling would also loosen the stride, but the showcase ships

Table 22: First walk-level read of the two shipped engines. 40 held-out sessions, 5-track seed, 20 autoregressive steps each; **cap** is the per-artist limit inside the journey. W_1 and $|\text{median step} - 0.261|$ lower-better; **vibe**, in-band share, artists, genres and **ground** higher-better; **drift** nearer zero better. The last row is the reference the metrics are calibrated against (its in-band share is 80 % by construction). Best cell per column bolded among the model arms. *Source: tools/walk_eval.py.*

Arm	W_1	med. step	in-band	vibe	drift	artists	genres	ground
GRU +anchor 0.4, cap 1 (shipped)	0.320	0.661	80.6 %	0.609	-0.089	20.0	9.93	0.345
GRU +anchor 0.4, cap 3	0.341	0.681	76.7 %	0.610	-0.075	13.2	6.53	0.393
GRU no anchor, cap 1	0.263	0.574	85.2 %	0.302	-0.277	20.0	8.78	0.276
Champion ($R'+M+C'$), cap 1	0.078	0.317	90.3 %	0.187	-0.186	20.0	10.33	0.289
Champion ($R'+M+C'$), cap 3	0.147	0.452	84.8 %	0.193	-0.224	11.1	5.33	0.306
<i>the user's real sessions</i>	0.000	0.261	80.0 %	—	—	—	—	—

permalinks, and “same recipe $\Rightarrow$ same journey” is now a promise, so a stochastic decoder was ruled out rather than untried.

CAP. A *hard* per-artist limit applied to the ordering (`seq_common.artist_cap_order`), after MMR, in both `eval_from_scores` and `rank_topk/predict_ranking`, so evaluation and serving agree by construction. Best-effort rather than absolute: where a prefix’s admissible candidate space is exhausted the remaining slots are backfilled with deferred items instead of returning fewer than k . Default `None`, so every pre-existing run and model is byte-identical; and because it needs no sonic vectors it is the only diversity lever here that ports to a thin client unchanged.

Why a cap and not a penalty — the two refuted controls. Both soft anti-eager levers of Section 15 fail for one shared reason: a soft penalty in score space loses to a steep relevance gradient. `eager_beta`, the training-time regularizer, is *mis-specified*: it hinges on the absolute value of $\cos(\text{prediction}, \text{current track})$, which a hedged InfoNCE prediction never drives above ~ 0.6 , whereas eagerness is a *ranking* property — whether same-artist items occupy the top slots — not a property of an absolute similarity. `mmr_lambda`, the eval-time re-rank, was additionally *eval-only* until the serving path was fixed, so the λ frontier had never been a deployable lever at all; and even fixed it measures inert on the pathological case. On the real playlist **rock chabon ahre** the model ranks 59 of its top 60 candidates as one artist and the next artist first appears at **rank** 75, while MMR’s entire diversity budget there is worth $0.3 \times 0.26 \approx 0.078$ of normalized score — nowhere near enough to pay for a 70-rank relevance drop. A cap does not compete on score at all: it *cannot be outscored*, the same property `seq_extend`’s `artist_cooldown` already relied on. The showcase’s own soft artist penalty turned out to be effectively **binary** in the same way — at 0 it produced artist blocks with runs up to 13, and at *any* value ≥ 0.25 every 30-step walk returned 30 distinct artists, max run 1, concentration 0.000 — so the slider could not reach the middle at all, and a hard cap exposed as a journey shape (Explore / Balanced / Continue / Blocks) is what reaches it.

The cap on the one-shot surface, for the record. Table 23 is the cap measured where the rest of this document measures: the canonical split’s one-shot top-10, paired on bit-identical cached scores. It cuts `artist_conc@10` from 0.692 to 0.123 at cap 3 and raises the generation-3 `holisticness@10` from 0.008603 to 0.033862, at a cost of 73 of 1431 exact hits. The result that was

not predicted is that **music@10 goes up, CI above zero, at every cap on both models tested** (+0.011 to +0.023): the pre-registered failure mode — de-eagering by degrading into generic filler — does not occur, because the items promoted from rank 75 and beyond are *more* musically apt to the true continuation than the same-artist tracks they displace. The cap trades *literal* relevance for *graded* relevance. On the champion plus **markov_gate** (a serving-time gate that suppresses the Markov leg where the current item has no bigram support; its own campaign, dated 2026-07-27c, is not yet consolidated into this synthesis) the same cap 3 gives **artist_conc** 0.598→0.113 and **holisticness@10** 0.018699→0.048063, reproduced to every digit by three registered confirming runs (run-20260730-235524-0ec1d-seq-blend at cap off, run-20260730-235524-93d38-seq-blend at cap 3, run-20260730-234635-a6252-seq-blend at cap 5), and it **hardens on the disjoint 0.64-cold split seq-20260718-211238** on two pre-registered legs: Δ **artist_conc** -0.52699 [-0.54546, -0.50825] CI below zero *and* Δ **music@10** +0.02644 [+0.01919, +0.03368] CI above zero, with the relevance gain *larger* on the colder window. Two cautions travel with it: exact recall is the only regressing metric at every cap, and the canonical split’s absolute **music@10** ≥ 0.43 floor does **not** transfer — the second split’s own control sits at 0.4295 on a legitimately colder window, so floors must be derived per baseline and pre-registration must be on the paired Δ .

Table 23: The hard per-artist cap on the one-shot surface, on the deployed champion ($R'+M+C'$, ungated), canonical split seq-20260715-131139, $n = 1431$. Off-run measurement: scores are computed once per model and memoized, so every cap arm sees bit-identical score vectors and the pairing is exact rather than merely matched (2000 resamples, rng 1337). **holisticness@10** is generation 3 (Section 18.2). **artist_conc**, **artist_adj** lower-better; the rest higher-better. *Source: 2026-07-30c-artist-cap-measurement.md.*

cap	R@10	hits	MRR	artist_conc	artist_adj	ild	music@10	H@10 (gen-3)
off	0.211740	303	0.121985	0.691653	0.609713	0.406374	0.453790	0.008603
5	0.178896	256	0.119693	0.252613	0.337247	0.563417	0.474291	0.027258
3	0.160727	230	0.116915	0.122975	0.215304	0.613641	0.476787	0.033862
2	0.138365	198	0.112313	0.061946	0.150384	0.639717	0.474317	0.036331

The anchor $\times$ stride frontier. With the two score-space controls in hand the question becomes whether the *conjunction* is reachable at all: a real-listening stride *and* a held vibe. Table 24 sweeps 24 cells — 16 on the GRU, 8 on the champion, all at cap 1 — against a target window pre-registered before the sweep: median step inside the real band’s middle (0.20 to 0.38) *and* **vibe** ≥ 0.50 . Exactly two cells land inside it, one per engine (Figure 15). The structure of the grid is the finding: the two controls are genuinely orthogonal — along a fixed anchor, raising the stride walks the median step down monotonically while the vibe decays much more slowly — so the window is reachable only with both on, and neither shipped configuration was inside it. The champion’s anchor also has to be applied in the *raw* PCA-192 space rather than its own projected geometry, which is why the frontier reaches 0.503 where the first probe capped at 0.44.

19.4 The matched-sample head-to-head

Rationale. The two sweeps above each found a cell inside the window, but on different samples, and W_1 cannot be paired — so the finalists had to be re-run against one another on *identical* material before anything could be claimed. **tools/walk_headtohead.py** walks five arms over the same 40 held-out sessions from the same 5-track seeds, at cap 1, and reports paired bootstrap differences

Table 24: The anchor (a) $\times$ stride (s) frontier, 20 held-out sessions $\times$ 20 steps, cap 1, both engines. W_1 lower-better; “med. step” is compared against the user’s real 0.261; **vibe**, genres, **ground** higher-better; **drift** nearer zero better. The window is median step $\in [0.20, 0.38]$ and **vibe** ≥ 0.50 , pre-registered before the sweep; the two rows that satisfy it are bolded. The shipped GRU configuration is $a = 0.4$, $s = 0$; the champion as deployed is $a = 0$, $s = 0$. *Source: tools/walk_frontier.py.*

Engine	a / s	W_1	med. step	vibe	drift	genres	ground
GRU	0.0 / 0.0	0.287	0.607	0.286	−0.328	9.70	0.270
GRU	0.0 / 0.3	0.273	−0.018	0.200	−0.157	8.95	0.291
GRU	0.0 / 0.6	0.564	−0.374	0.123	−0.102	12.00	0.244
GRU	0.0 / 1.0	0.702	−0.557	0.084	−0.036	15.80	0.224
GRU	0.2 / 0.0	0.323	0.663	0.563	−0.124	9.40	0.345
GRU	0.2 / 0.3	0.186	0.094	0.426	−0.098	11.05	0.335
GRU	0.2 / 0.6	0.473	−0.251	0.330	−0.067	13.25	0.303
GRU	0.2 / 1.0	0.675	−0.522	0.185	−0.077	15.05	0.253
GRU	0.4 / 0.0 (<i>shipped</i>)	0.361	0.709	0.639	−0.090	9.95	0.349
GRU	0.4 / 0.3	0.104	0.297	0.588	−0.076	11.05	0.358
GRU	0.4 / 0.6	0.356	−0.127	0.437	−0.044	12.55	0.303
GRU	0.4 / 1.0	0.588	−0.405	0.302	−0.027	15.20	0.287
GRU	0.8 / 0.0	0.364	0.713	0.678	−0.067	11.00	0.352
GRU	0.8 / 0.3	0.190	0.506	0.672	−0.070	12.15	0.351
GRU	0.8 / 0.6	0.122	0.176	0.605	−0.038	12.60	0.358
GRU	0.8 / 1.0	0.420	−0.188	0.460	−0.016	14.65	0.335
Champion	0.0 / 0.0 (<i>deployed</i>)	0.062	0.318	0.185	−0.203	11.00	0.275
Champion	0.0 / 0.5	0.240	−0.032	0.104	+0.002	12.15	0.266
Champion	1.0 / 0.0	0.239	0.577	0.551	−0.138	8.95	0.327
Champion	1.0 / 0.5	0.072	0.282	0.503	−0.130	9.55	0.324
Champion	2.0 / 0.0	0.276	0.596	0.625	−0.060	8.90	0.359
Champion	2.0 / 0.5	0.170	0.472	0.615	−0.105	9.45	0.338
Champion	4.0 / 0.0	0.300	0.645	0.669	−0.051	9.85	0.338
Champion	4.0 / 0.5	0.236	0.544	0.664	−0.085	9.70	0.340

(2000 resamples, rng 1337) against the **shipped** configuration — the single GRU at anchor 0.4, stride 0 — on five per-session metrics: **vibe** (higher better), **stride_err** (lower better), **drift** (nearer zero), **genres** (higher) and **ground** (higher).

Results. Table 25 gives the means and Table 26 the paired differences; Figure 16 plots the latter one panel per metric, each on its own scale.

Interpretation. Read down the **stride_err** column first: all four tuned arms cut it by ~ 0.25 with the CI below zero, which is simply the stride control working — the shipped engine’s 0.399 error was $\sim 2.6\times$ the next worst arm’s. The interesting variation is what each arm *pays* for it. Pushing the single GRU to a real stride **costs vibe** (−0.0409, CI below zero) and the champion pays far more (−0.1327, CI below zero): for these two engines stride and vibe are in tension, exactly as the frontier’s gentle-but-real slope predicted. Both dual-tower arms escape the trade-off. **latent/latent** at $a0.8\ s0.4$ *gains* vibe (+0.0297, CI above zero) and genres (+1.05, CI above zero) while matching the stride; **latent/cummean** at $a0.8\ s0.5$ holds vibe (+0.0013, straddling — held, not traded) and is the only arm with *three* wins, adding 1.5 genres (CI above zero) and the only significant drift improvement (+0.0351, CI above zero, i.e. the journey ends closer to where it started). On the objective as stated — mood-holding, genre-exploring sequences — **latent/cummean** is the winner, and the single GRU that was actually shipped is not on the frontier at all.

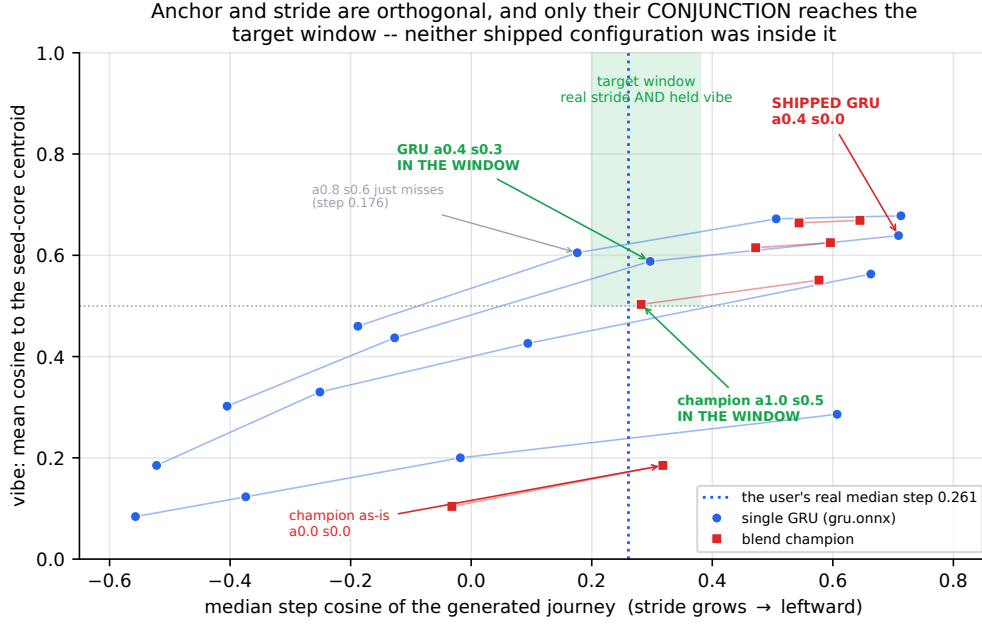


Figure 15: The two retrieval-time controls are orthogonal, and only their *conjunction* reaches the target window (shaded: median step $\in [0.20, 0.38]$ and $\text{vibe} \geq 0.50$). Each line joins a fixed anchor as the stride grows, which moves the journey leftward (larger steps) while the vibe decays slowly — so the trade-off is real but gentle. Neither shipped configuration is inside the window: the GRU at $a = 0.4$, $s = 0$ holds a vibe of 0.639 but strides 0.709, and the champion as deployed strides 0.318 at a vibe of 0.185. 20 held-out sessions $\times$ 20 steps, cap 1 (`tools/walk_frontier.py`).

19.5 The dual-tower reversal, and what the record already knew

This is worth stating plainly, because it is the most consequential methodological result in this document: **the dual-tower family was closed by three campaigns on recall grounds and is the winner on the stated objective.** Section 16 refuted it across the complete $\{\text{latent}, \text{delta}, \text{cummean}\}^2$ view matrix; the `cummean/cummean` arm was even flagged *degenerate* for topping the generation-1 holisticness at Recall@10 0.066. Every one of those judgements was made on one-shot next-track accuracy, and none of them is wrong on that surface — the reversal is not a correction of the recall verdicts but a demonstration that they answered a different question. Section 18.1’s power analysis makes the two readings compatible rather than contradictory: what the dual-tower campaign established was the absence of a *large* recall gain, which is fully consistent with the family being better at something the recall metric does not measure.

The corroboration was already in the record, filed as a negative. The interpretability note at `predictors/seq_dualgru.py:113` records that the `cummean` tower “added real linear directions (CKA 0.458) but **zero** next-item concepts tower A lacked.” Read as a next-track result that is dead weight, and it was written down as such. Read on the walk surface it is the useful part: a tower whose content is orthogonal to next-item prediction is exactly a tower carrying *session state* rather than adjacency. Two structural facts complete the explanation. The family retrieves in the *raw* item-latent space with no learned projection, so a raw-space mood anchor is native to its geometry — unlike the champion, whose InfoNCE projection strips precisely the sound-similarity structure an anchor needs. And one promoted arm reads the *causal mean* of the prefix, which is structurally a running-average signal.

Table 25: Walk-metric means over the 40 identical held-out sessions (20 steps, cap 1). **stride_err** = $|\text{median step} - 0.261|$ (lower better); **vibe**, **genres**, **ground** higher-better; **drift** nearer zero better. Best cell per column bolded. *Source: tools/walk_headtohead.py.*

Arm	vibe	stride_err	drift	genres	ground
GRU shipped, $a0.4\ s0.0$	0.609	0.399	-0.089	9.925	0.345
GRU tuned, $a0.4\ s0.3$	0.568	0.129	-0.080	10.450	0.357
dual latent/latent f0, $a0.8\ s0.4$	0.638	0.157	-0.063	10.975	0.351
dual latent/cummean f0 , $a0.8\ s0.5$	0.610	0.144	-0.053	11.425	0.351
Champion ($R'+M+C'$), $a1.0\ s0.5$	0.476	0.135	-0.131	9.600	0.334

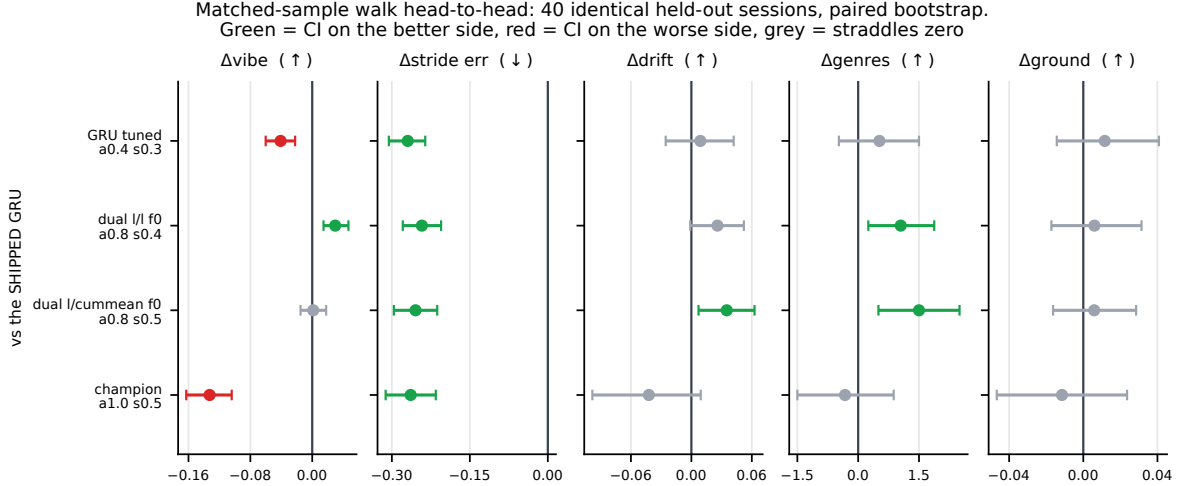


Figure 16: The matched-sample head-to-head, one panel per metric on its own scale — deliberately not a dual axis, which would imply a comparability these five quantities do not have. Green is a CI entirely on the metric’s better side, red entirely on the worse side, grey straddles zero. Every arm buys the real stride; only the dual-tower arms buy it without paying vibe, and **ground** straddles for *every* arm.

One sub-hypothesis is refuted. The natural prediction from the above — that the **cummean** tower holds a vibe *natively* — is false: with both controls off its vibe is 0.236, statistically indistinguishable from **latent/latent**’s 0.234. The architecture does not supply the mood hold on its own; it only becomes the best host for one once the anchor and stride are applied. That is a weaker and more interesting claim than the one the structural argument suggested, and it is the one the measurement supports.

19.6 Deployment, and what is not established

The tuned dual-tower arm is now the **default engine** in the showcase, exported as **dualgru.onnx** (3.16 MB) with ONNX-versus-torch parity of 1×10^{-7} across every prefix length $T = 1 \dots 40$ — the **cummean** cumulative sum was the trace risk, and it is the reason the parity check swept T rather than spot-checking it. The single GRU is kept alongside for comparison, and the tuned anchor/stride live in the deployment manifest rather than in code, so re-tuning is a re-bake and not a code edit. The blend champion was removed from the showcase end-to-end: served from a Worker its walk cost ~ 0.94 s of CPU per batch of 8, roughly $100\times$ the 10 ms CPU cap of the

Table 26: Paired per-session differences against the **shipped** GRU (*a0.4 s0.0*) over the same 40 sessions, 2000 bootstrap resamples, rng 1337. “Verdict” reads the CI relative to the metric’s better direction: **stride_err** is lower-better, so a CI below zero is a *win*; **drift** starts negative for every arm, so a positive Δ moves it toward zero. Wins bolded. *Source: tools/walk_headtohead.py.*

Arm	Metric	Δ [95% CI]	Verdict
GRU tuned <i>a0.4 s0.3</i>	vibe	−0.0409 [−0.0602, −0.0221]	CI<0: <i>costs vibe</i>
	stride_err	−0.2695 [− 0.3058 , − 0.2358]	CI<0: win
	drift	+0.0089 [−0.0255, +0.0420]	straddles
	genres	+0.525 [−0.475, +1.500]	straddles
	ground	+0.0116 [−0.0143, +0.0408]	straddles
dual latent/latent f0 <i>a0.8 s0.4</i>	vibe	+0.0297 [+0.0146 , +0.0468]	CI>0: win
	stride_err	−0.2423 [− 0.2790 , − 0.2053]	CI<0: win
	drift	+0.0259 [−0.0011, +0.0521]	straddles
	genres	+1.050 [+0.250 , +1.876]	CI>0: win
	ground	+0.0060 [−0.0172, +0.0314]	straddles
dual latent/cummean f0 <i>a0.8 s0.5</i>	vibe	+0.0013 [−0.0151, +0.0180]	straddles: <i>held</i>
	stride_err	−0.2545 [− 0.2962 , − 0.2127]	CI<0: win
	drift	+0.0351 [+0.0071 , +0.0627]	CI>0: win
	genres	+1.500 [+0.500 , +2.500]	CI>0: win
	ground	+0.0059 [−0.0163, +0.0285]	straddles
Champion ($R'+M+C'$) <i>a1.0 s0.5</i>	vibe	−0.1327 [−0.1629, −0.1041]	CI<0: <i>costs vibe</i>
	stride_err	−0.2638 [− 0.3120 , − 0.2155]	CI<0: win
	drift	−0.0422 [−0.0983, +0.0093]	straddles
	genres	−0.325 [−1.500, +0.876]	straddles
	ground	−0.0115 [−0.0467, +0.0236]	straddles

free plan the showcase is deployed on, and batching cannot rescue a single step that already costs ~ 117 ms.

What this section does *not* establish. Four limits, stated plainly. **(1)** ground straddles zero for *every* arm (Table 26), so no arm is established as better at predicting the real continuation — the wins are about journey *shape* only, and the grounding guard-rail confirms only that no arm degraded into pleasant irrelevance. **(2)** One split, one seed, 40 sessions of 20 steps; the walk metrics have had no second-split hardening of the kind Section 12.1 applies to the recall crown, and the target window itself was chosen from the first sweep’s spread. **(3)** Nothing here has been *listened* to. Every claim is a geometric claim about cosines in a PCA-192 space, and “holds a mood” is operationalized as “stays near a seed centroid,” which is a hypothesis about perception, not a measurement of it. **(4)** The project’s primary metric is **unchanged**: holisticness@10 remains `[metrics].primary` and this document’s board remains denominated in Recall@10 (Section 18.2). Whether the walk metrics should become primary is an open decision, not a settled one — and until it is settled, the honest summary is that the walk surface is now *measurable*, and on it one refuted family wins. *Sources: 2026-07-30c-artist-cap-measurement.md, PROJECT-FACTS.md (roll-up 2026-07-31b), tools/walk_eval.py, tools/walk_frontier.py, tools/walk_dualgru.py, tools/walk_headtohead.py.*

20 Conclusions

Fifteen campaigns (Jul 2026) on the next-track retrieval task converge on a clear story. (1) A session sequence model, even at its best (GRU-infonce), only *ties* the first-order Markov co-listening bar on Recall@10 and loses on MRR: predicting a next-item latent and retrieving by one cosine hop is a harder route to the exact next track than a bigram that directly memorizes adjacency. (2) The training *loss* is the dominant single-model lever (InfoNCE $\approx 2.3\times$ cosine); network *topology* — depth, width, residual, direction — is a spent lever, and no architecture beats the bar. (3) The breakthrough is *rank fusion*: an R+M z-blend of the GRU and Markov legs beats the bar by $\sim +61\%$ Recall@10 (three-seed confirmed), because the two legs rank *different* correct tracks. The win is about signal *diversity*, not model count — adding a redundant neural leg does not help, and a learned XGB stacker *underperforms* the fixed z-blend, because its pointwise training does not capture the ranking objective. (4) An item-representation sweep found latent *dimension*, not the compressor, to be the lever: linear PCA beats the nonlinear autoencoder at every $\text{dim} \geq 64$ (the acoustic-preservation hypothesis is refuted), the curve peaks at PCA-192 and turns over by 256, and PCA-128 is the confirmed single-model base. (5) On the rich PCA space the one genuinely additive new signal — *content-kNN*, which only *tied* the champion on the AE-64 space — becomes a win: a three-leg R+M+C z-blend on PCA-192 is the current best-on-record row at Recall@10 0.1859, MRR 0.1025 (paired Δ over the prior champion +0.0133, CI above zero, confirmed across five reproductions). (6) Two honest caveats bound the crown: the reproduction confirm had zero spread because **seq-nexttrack** is deterministic on this CPU (so it verified determinism, not seed-robustness), and the paired- Δ lower bound clears zero by only +0.0007 on a single split — hardening it needs a second leak-free test split. (7) Every ranking predictor here is train-only and off-server, so the crown is a best-on-record *row*, not a promoted, app-exportable model; the prior champion R+M on AE-64 remains the productization anchor, and registering a three-leg blend with a **predict** subcommand is the standing top follow-up. (8) A subsequent literature-fit campaign then *superseded* the crown with the first genuinely-additive new signal since content: a supervised *learned content projection* ($R'+M+C'$), an InfoNCE metric map of the PCA-192 item space fit toward next-track adjacency in which both the GRU and content legs retrieve, at Recall@10 0.2117, MRR 0.1220 (paired Δ over R+M+C +0.0259, CI>0). It resolves both R+M+C caveats — it is seed-robust across three seeds and cold-reaching (its content leg scores cold targets as well as warm) — and it was *registered, promoted and servable* (**blend-gru-markov-content-proj**, best-models rank 1), closing the productization gap; an objective / rank / τ scan confirms InfoNCE $\gg$ BPR and holds the default, and the crown then hardened on a second disjoint 0.64-cold split (paired Δ +0.0426, CI>0, the margin *wider* on the colder split). Every borrowed literature-fit method (session-kNN, rank-fusion / learned combiners, GRU-side loss tricks, Markov smoothing) closed — the only additive gain came from new signal, not smarter fusion. (9) An item-vector representation exploration then confirmed *why* the projection is the champion: the equal-weighted PCA-192 geometry is a real handicap for a raw model (six of eight whitened / acoustic-dropped / metric spaces lift the frozen GRU, paired Δ CI>0), but the gain is fully *absorbed* by the learned projection (no space beats the champion on readout A) — whitening and the projection are the same fix, and the champion has already pulled it. (10) Finally, a continuous “sounds-alike” metric, *music@10*, was introduced and back-filled across all 53 runs: it correlates with Recall@10 ($r = 0.609$) but crowns a different winner (the content-regressing **seq-nexttrack** lands the musically-closest misses, best 0.482), corroborating the exact-recall findings without overturning them; it is display-only and the primary metric stays Recall@10. (11) Finally, a measurement campaign added a display-only *session-holisticness* suite (artist_adj@10 / mood_coh@10 / ild@10) and two anti-eager levers. It

found the eagerness symptom *tracks* Recall@10 inside the blend family — the champion is the most artist-eager model (artist_adj@10 0.610) — but the eagerness is sourced from the *Markov* bigram leg, not the RNN (single GRU / LSTM artist_adj@10 $\approx$ 0.345), and the GRU and LSTM are statistically tied on both recall and eagerness (the “GRU wins because more eager” hypothesis is refuted). Of the two levers the eval-time MMR re-rank is the strong knob ($\lambda=0.9$ near-free: recall tie, ild@10 0.406 $\rightarrow$ 0.436; $\lambda \approx 0.7$ de-eagers for ~ 2 recall points; music@10 even rises as the list diversifies) while the training-time β regularizer is weak (β 0 to 0.2 moves artist_adj@10 only -0.0074); nothing was promoted, and $\lambda \approx 0.9$ is flagged for a future user holistiness decision (Section 15). (12) Two final architecture campaigns then closed the last open architecture axes, and both are nulls — but nulls with mechanism. A jointly-trained *dual-tower* GRU, crossing the complete `{latent,delta,cummean}`² view matrix against a 0.098% parameter-matched single tower, fails to beat that tower on any of eight arms (Section 16). This is the *fourth* independent learned-combiner null and the one that matters most, because it answers the standing rebuttal to the earlier three (“of course a bolted-on combiner fails — train the fusion end-to-end”): trained end-to-end at the representation level, it still fails. The mechanism is that the raw `latent` view already *subsumes* the others — a GRU over raw latents integrates movement and running mean internally, and the one genuinely complementary pair is `delta+cummean`, which beats both its parents yet is hopeless in absolute terms. Two by-products are durable: the MLP fusion head is actively *harmful* (removing it wins in every regime measured, three times over, Table 19), and single-GRU width is now closed on PCA-192 as it already was on AE-64. A learned per-step *pre-encoder* then fails harder (Section 17): the decisive single-tower ablation *loses* outright (-0.0140 , CI below zero, 156 hits against 176), the dose curve is flat to twice the input dimension, and placement is irrelevant. Because a *linear* pre-encoder is provably expressivity-neutral here, the rectifier is the only candidate cause — and a counterfactual interpretability read confirmed it, measuring the affine map neutral at $\sim 0.8\sigma$ and the ReLU costing $\sim 5.9\sigma$, with `frac_exact_zero` 0.5327 matching `frac_negative_coords` 0.5327 to four digits: half-wave rectification of a zero-centred latent deletes exactly its negative half. Across five architecture nulls the pattern is now hard to dismiss — on this single-user, 55%-cold corpus, adding representation machinery *around* a single GRU over a good latent space does not help, while every real gain came from improving the latent space itself or from the fixed z-blend. (13) Finally, two facts about the *instrument* rather than any model reframe the whole architecture arc (Section 18). The nulls are **power-limited**: at $n = 1431$ the paired-bootstrap CI half-width is 0.0124 on Recall@10 and ~ 0.0018 on holistiness@10, while a whole architecture family spans only 0.028 and 0.0031 — so of eight adjacent arms roughly two tiers are separable on recall and one on holistiness, and resolving a 0.005 difference would need ~ 8800 sessions. The honest reading is therefore that *large harms were detected reliably while moderate wins were never detectable at all*, which is a different claim from “these architectures do not work” and yields a prioritization rule: further architecture arms are a poor use of runs on this split whatever they return, because the axis’s whole dynamic range is barely twice the measurement error. And *holistiness@10* exists in three mutually non-comparable code generations (3-factor; 4-factor grounded; grounded plus artist-concentration), whose differences dwarf every model difference and under all of which the servable champion ranks *last* — so this synthesis keeps its board denominated in Recall@10 and tags every holistiness column with its generation (Table 21). (14) Finally, a campaign about the *surface* rather than the model finds that the board above had been scoring the wrong one (Section 19). Everything else in this document is a one-shot top-10, while the shipped product serves an autoregressive *walk* — and *holistiness@10*’s `mood_coh` anchors to the *prefix* centroid, which on a walk grows with the journey, so a run of individually legal steps that ends in the opposite region of the space scores well at every step: drift was invisible

by construction, and the composite is $5.5\times$ more sensitive to relevance than to eagerness besides. Calibrated against the user’s own 73 632 consecutive transitions (median step cosine 0.261, 27.0 % negative), the engine the showcase shipped walks $2.5\times$ too tight (median step 0.661) — “flat despite lots of artists” was many small safe moves — while the blend champion matches the real stride but cannot hold a mood, because its InfoNCE projection encodes what *follows* rather than what *sounds alike*. Three deterministic retrieval-time controls fix the reachability: a seed anchor, a stride term (deterministic on purpose — sampling would break the permalink guarantee), and a *hard* per-artist cap, which succeeds exactly where the two soft anti-eager levers failed because it cannot be outscored (on the pathological playlist 59 of the top 60 candidates are one artist and the next artist sits at rank 75, against an MMR diversity budget worth ≈ 0.078 of normalized score; `eager_beta` is mis-specified on an absolute cosine when eagerness is a *ranking* property). The cap hardens on two disjoint splits and, unexpectedly, *raises music@10* at every setting: it trades literal for graded relevance. And on 40 identical held-out sessions under paired bootstrap the **dual-tower family, closed by three campaigns on recall grounds, is the winner on the stated objective** — `latent/cummean` at anchor 0.8 stride 0.5 takes stride error -0.2545 , genres $+1.50$ and drift $+0.0351$ with every CI clear of zero while *holding* vibe, where the single GRU (-0.0409) and the champion (-0.1327) both pay vibe for the same stride. The corroboration was already in the record as a *negative* (`seq_dualgru.py:113`: the `cummean` tower “added real linear directions (CKA 0.458) but zero next-item concepts tower A lacked”) — dead weight for next-track prediction is exactly session state for mood-holding, though the stronger reading is refuted: `cummean` does *not* hold a vibe natively (0.236 against 0.234), it only hosts one once the controls are applied. It is deployed as the showcase’s default engine; and `ground` straddles zero for every arm, so what is established is journey *shape*, not next-track truth, on one split, one seed and 40 sessions, with nothing yet listened to and the primary metric unchanged.

21 Source reports

This synthesis consolidates fifteen next-track campaign reports from `experiments/`:

- `2026-07-13-nexttrack-sequence-sweep.md` — the founding next-track campaign: a GRU/LSTM sequence sweep against the first-order Markov bar; the best GRU-infonce ties on R@10 but loses on MRR (Section 4).
- `2026-07-13-nexttrack-topology-scan.md` — the depth / width / residual / bidirectional topology scan; architecture is a spent lever, all arms cap below the Markov bar (Section 5).
- `2026-07-14-nexttrack-blend-and-ensemble.md` — the $\text{GRU} \times \text{Markov R+M z-blend}$ beats the bar $+61\%$ R@10 (the first next-track champion); no richer ensemble improves on it and the learned XGB stacker underperforms (Section 6).
- `2026-07-14-nexttrack-graded-relevance-and-new-legs.md` — first-class graded-relevance metrics (`artist@k` / `genre@k` / `artist_mrr`) and new leak-free base learners; content-kNN is the only additive signal, tying the champion on AE-64 (Section 7).
- `2026-07-15-nexttrack-compression-representation-bakeoff.md` — the item-representation compression bake-off (PCA vs autoencoder $\times$ latent dim 32/64/128) under a frozen GRU readout; PCA-128 is the single-GRU winner but the win does not carry into the blend (Section 8).

- 2026-07-15b-nexttrack-compression-dim-extension.md — the PCA/AE dimension sweep extended to 192/256; the curve peaks at PCA-192 and turns over, so PCA-128 stays the Phase-2 base (Section 9).
- 2026-07-15c-nexttrack-phase2-model-sweep.md — the model-family sweep on the rich PCA spaces; the fixed z-blend beats the learned stacker, LSTM is the best single model, and R+M+C on PCA-192 is crowned the new best-on-record row (Section 10).
- 2026-07-18-nexttrack-literature-fit-campaign.md — a bibliography-driven exploration; a supervised learned content projection ($R'+M+C'$) is the new best-on-record row (R@10 0.2117, paired Δ over R+M+C +0.0259 CI>0, seed-robust, cold-reaching), and every borrowed method (session-kNN, rank fusion, learned combiner, Markov smoothing, GRU importance negatives) closes (Section 11).
- 2026-07-18b-nexttrack-projection-registration-and-scan.md — the projection reproduced on-server, registered + promoted + servable as **blend-gru-markov-content-proj** (best-models rank 1); an objective / rank / τ scan (InfoNCE $\gg$ BPR; default held) and second-split hardening on a disjoint 0.64-cold split (paired Δ +0.0426 CI>0) (Section 12).
- 2026-07-18c-nexttrack-representation-exploration.md — an item-vector representation audit; the equal-weighted PCA-192 geometry handicaps the raw GRU (six of eight alt spaces lift it CI>0) but the gain is absorbed by the projection (no space beats the champion on readout A) (Section 13).
- 2026-07-20-musical-distance-metric.md — the continuous “sounds-alike” music@10 metric and its 53-run offline backfill; it correlates with Recall@10 ($r = 0.609$) but crowns a different winner, the content-regressing **seq-nexttrack** (Section 14).
- 2026-07-22-session-holisticness-and-antieager-levers.md — a display-only session-holisticness metric suite (artist_adj@10 / mood_coh@10 / ild@10) and two anti-eager levers (eval-time MMR **mmr_lambda**, training-time **eager_beta**); the champion is the most artist-eager model, the eagerness is sourced from the Markov leg, $\text{GRU} \approx \text{LSTM}$, and MMR λ is the strong knob (Section 15).
- 2026-07-25-nexttrack-dual-tower-fusion-scan.md — the dual-tower view-pairing $\times$ fusion-depth scan against a 0.098% parameter-matched single tower; **REFUTED** (the fourth learned-combiner null, and the one that closes the “train the fusion end-to-end” rebuttal), with the raw-latent view shown to subsume **delta/cummean**, the MLP fusion head shown actively harmful, single-GRU width closed on PCA-192, and holisticness/recall shown anti-correlated across the batch (Section 16).
- 2026-07-26-nexttrack-pre-encoder-scan.md — the learned per-step pre-encoder scan (presence $\times$ placement $\times$ dose) with the decisive single-tower ablation; **REFUTED and actively harmful** (-0.0140 Recall@10, CI below zero), the rectifier identified analytically and then measured (affine map neutral at $\sim 0.8\sigma$, ReLU $\sim 5.9\sigma$, **frac_exact_zero** = **frac_negative_coords** = 0.5327), plus the measured power analysis and the three holisticness generations (Sections 17 and 18).
- 2026-07-30c-artist-cap-measurement.md — the *hard per-artist cap*, the first de-eagering lever that works, and it works because it cannot be outscored: **artist_conc@10** 0.692 $\rightarrow$ 0.123

with `music@10` *rising* (CI above zero) at every setting, hardened on a second disjoint 0.64-cold split, together with the refutations of the two soft levers it replaces (Section 19.3). The walk-level material of Section 19 — the surface mismatch, the four walk metrics, the anchor $\times$ stride frontier, the matched-sample head-to-head and the dual-tower reversal — is transcribed from the `PROJECT-FACTS.md` roll-up of 2026-07-31b and from the four measurement tools it cites (`tools/walk_eval.py`, `walk_frontier.py`, `walk_dualgru.py`, `walk_headtohead.py`), whose module docstrings are the authority for how each metric is computed; that material has no campaign report of its own yet.